

MINI-PROJECT-2

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Che 396: Senior Design I

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TASK 2: Process Design in Vinyl Chloride Polymer Production

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Executive Summary

This project develops and gives a minimum-utility Heat Exchanger Network (HEN) for a four-stream process via pinch analysis with a target $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$. We first established the thermodynamic basis two cold streams (Feed: $60 \rightarrow 220\text{ }^{\circ}\text{C}$; Reactant: $30 \rightarrow 180\text{ }^{\circ}\text{C}$) and two hot streams (Distillate: $160 \rightarrow 40\text{ }^{\circ}\text{C}$; Bottoms: $180 \rightarrow 40\text{ }^{\circ}\text{C}$) and calculated stream heat loads from given C_p values. Hot and cold composites were then constructed by temperature adjustment ($\pm 5\text{ }^{\circ}\text{C}$) to apply $\Delta T_{\min}$, providing the actual and adjusted composite curves to determine the pinch. The analysis determined pinch temperatures of $160\text{ }^{\circ}\text{C}$ (hot) and $150\text{ }^{\circ}\text{C}$ (cold) with a composite intersection at $155\text{ }^{\circ}\text{C}$, meeting the $10\text{ }^{\circ}\text{C}$ approach criterion. From these figures, we sized single responsibilities (kW) and verified that the combined representation gives a net-load balance at the pinch i.e., the most energy-effective foundation for heat recovery and the starting place for a viable HEN above and below the pinch. The task then translated the composite-curve targets into a line-diagram HEN: matches were selected to achieve maximum internal recovery without violating end-approach constraints, with counter-current configuration favored and stream segments allocated across the pinch based on established rules. This created a network that was compatible with the calculated pinch, composite-curve targets, and stream duties, and minimized purchased utilities. En route, we checked AI-derived first passes against manual calculation and lecture software (Excel cascade), removing common model errors (phantom streams, hot/cold misclassification, unit mixing, and sign errors) before finishing the network. The result is an open, auditable HEN design supported by numbers of the actual/shifted composite curves, the pinch calculation, and the finished HEN schematic, along with a utility summary matched to the composite-curve targets. In short, the analysis shows how disciplined pinch design in conjunction with strict verification of AI-generated designs leads to a physically

feasible network that meets $\Delta T_{\min}$, adheres to pinch principles, and achieves minimum utility consumption for the four-stream system in question.

Design Plan

The objective of this project is to develop a heat exchanger network (HEN) that reduces utility costs by maximizing internal heat recovery between process streams. The design aims to minimize the use of high-pressure steam and cooling water while keeping all temperature differences within feasible limits.

The design is based on known stream data, including inlet and outlet temperatures, mass flowrates, and heat capacities. A minimum temperature difference $\Delta T_{\min}$ of 10°C was used to ensure realistic heat transfer. The process is assumed to operate under steady-state conditions with no heat losses to the surroundings.

The pinch analysis method was used to guide the design. Stream data were analyzed to calculate energy changes using $Q = \dot{m}C_p\Delta T$. Composite curves were created to find the pinch temperature, which separates the system into “above pinch” (heating) and “below pinch” (cooling) regions. Stream matches were then selected to maximize heat recovery within these regions without transferring heat across the pinch.

The expected result of this design is a network that achieves significant energy savings and lower utility costs compared to a process without heat integration. The HEN layout developed from this plan is thermodynamically feasible and improves overall process efficiency.

Process Flow Diagrams

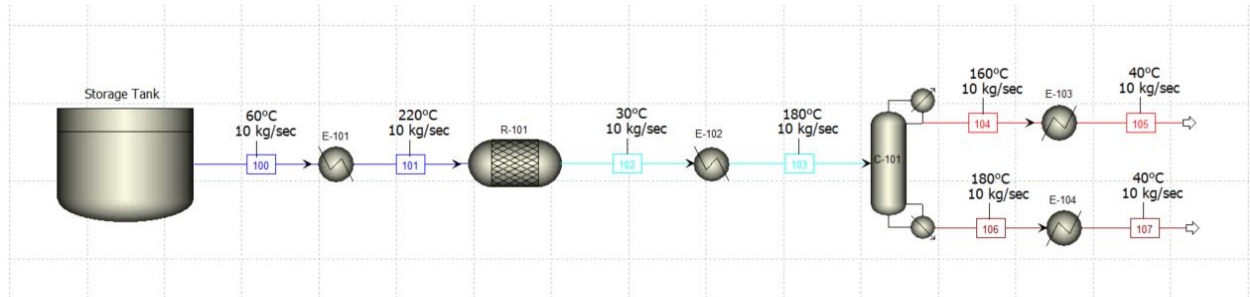


Figure 1: Process Flow Diagram without Heat Exchanger Design integration

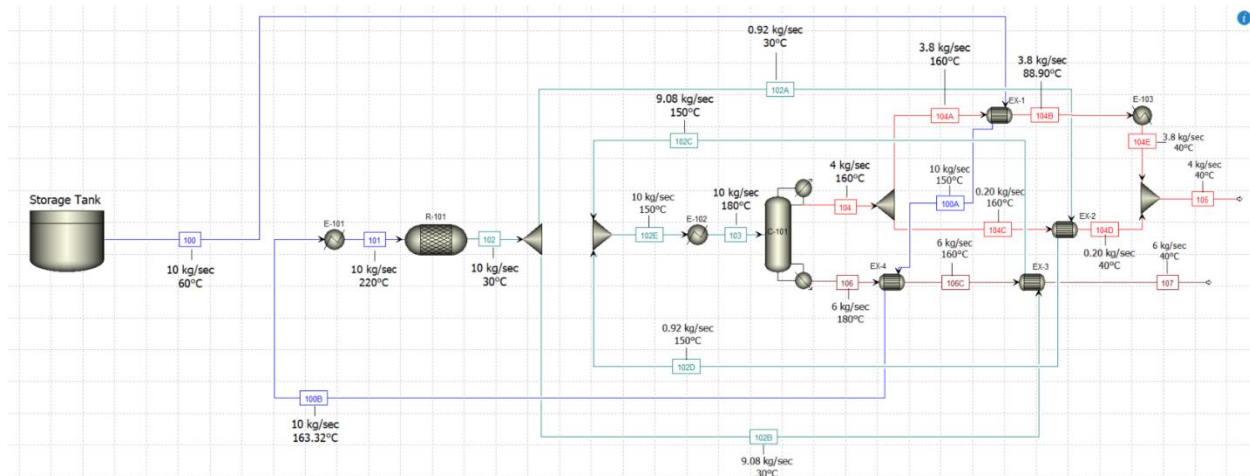


Figure 2: Process Flow Diagram with Heat Exchanger Network Design Integration

Composite Curves

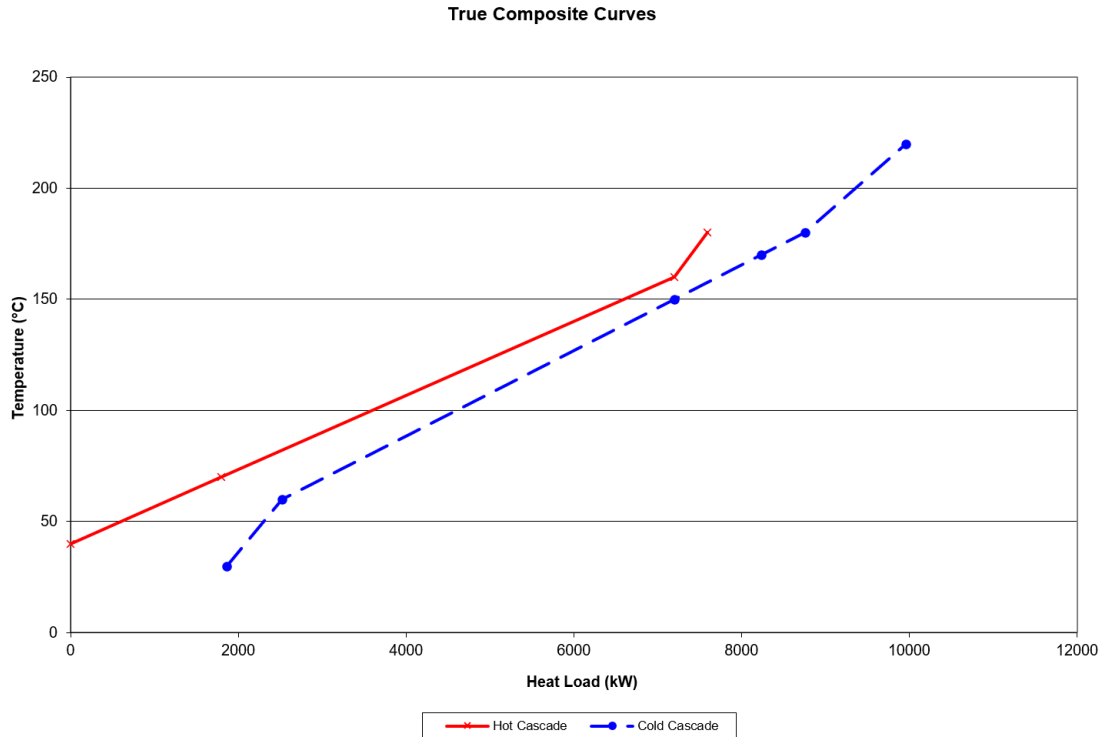


Figure 3: True Composite Curve

The composite curve was conducted under the assumption that there were four streams: Feed (cold), Reactant (cold), Distillate (hot), and Bottom (hot). Excel was used, but manual calculations were also done to supplement the results. $\Delta T_{\min}$ was 10 °C, divided by 2 to find the shift, which depending on the type of stream was either +5 °C or -5 °C. Using the Cp values and the differences between the supplied and target temperatures, the heat loads (kW) were found to be -4800, -3300, 4800, and 2792.2. Using this data, a Temperature vs. Heat Load graph was constructed. The graph consisted of two lines, the cold stream and the hot stream, which were plotted with their associated heats. The hot and cold pinch temperatures were found to be 160 °C and 150 °C, which agrees with $\Delta T_{\min}$. Taking the average, the pinch point occurs at 155 °C. This

ensures that the net load energy is 0 kW and represents the most optimal temperature for energy recovery.

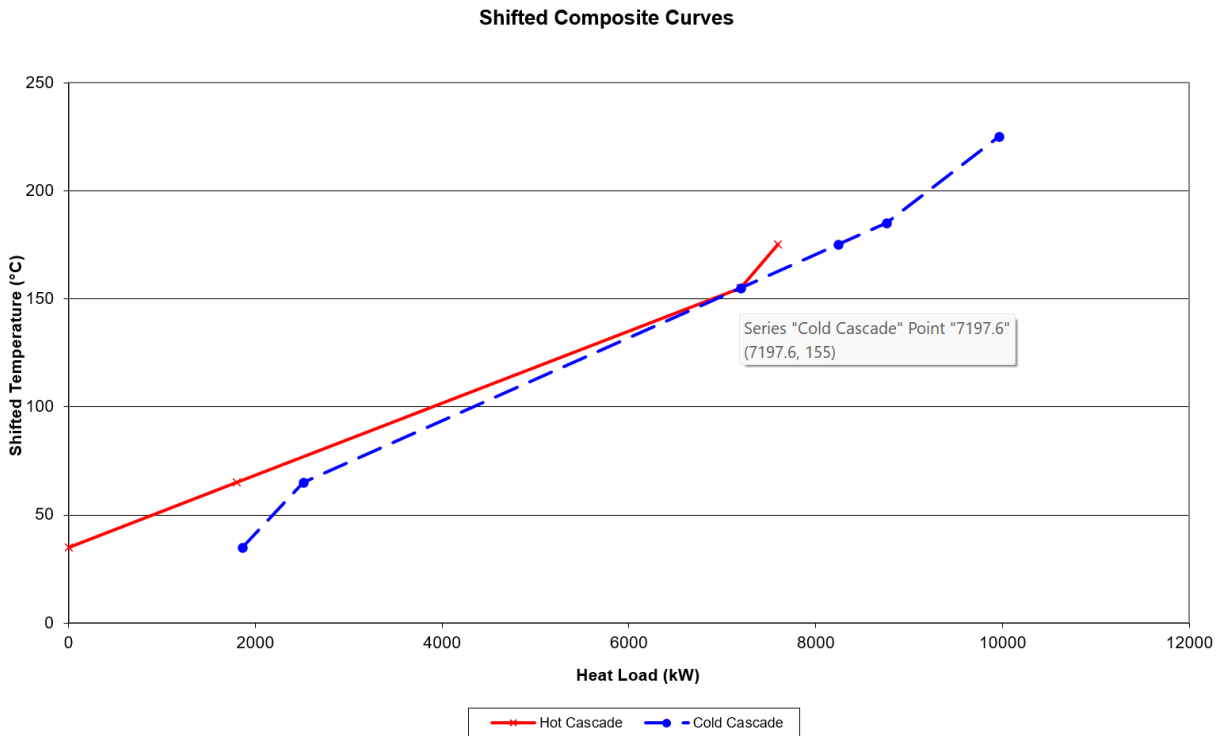


Figure 4: Shifted Composite Curve

$\Delta T_{\min}$ Calculation

The optimal $\Delta T_{\min}$ is given at 10 °C. If this value was not given in the first place, further analysis is required to determine $\Delta T_{\min}$. It can be determined by plotting capital cost, utility cost, and total annual cost against various $\Delta T_{\min}$ values. Smaller $\Delta T_{\min}$ values lead to greater heat recovery and lower utility costs but require larger heat exchangers, increasing capital cost. Conversely, larger $\Delta T_{\min}$ values reduce exchanger size but increase utility demand. By calculating these costs at several trial $\Delta T_{\min}$ values and graphing them, the capital cost curve will rise with smaller temperature differences, while the utility cost curve will fall. The sum of both, representing total cost, forms a U-shaped curve. The point where the total cost reaches its

minimum corresponds to the optimal $\Delta T_{\min}$, which provides the most economical balance between energy savings and equipment investment.

Heat Exchanger Network Design

To design the optimal heat exchanger network design, T_{pinch} , $T_{\text{pinch(cold)}}$, $T_{\text{pinch(hot)}}$ must be calculated. Using the excel file provided in lecture, the starting and target temperatures of all 4 streams must be uploaded to the file. The file automatically calculates the stream data once given the Heat Capacity. T_{pinch} is chosen as the interval temperature that sets enthalpy to 0. From Figure 5, $T_{\text{pinch}} = 155^\circ\text{C}$.

1. Minimum temperature approach

$\Delta T_{\min}$ 10 °C

2. Stream data

Stream No.	Actual temperature (°C)		Interval temperature (°C)		Heat capacity flow rate CP (kW/°C)	Heat load (kW)
	Source	Target	Source	Target		
1	60	220	65	225	30	4800
2	30	180	35	185	22	3300
3	160	40	155	35	40	4800
4	180	40	175	35	19.98	2797.2
5	0	0	5	5	0	0
6	0	0	5	5	0	0
7	0	0	5	5	0	0
8	0	0	5	5	0	0

3. Problem table

Interval	Interval temp (°C)	Interval ΔT (°C)	Sum CPc - sum CPh (kW/°C)	dH (kW)	Cascade (kW)	(kW)
	225				0	2360.4
1	185	40	30	1200	-1200	1160.4
2	175	10	52	520	-1720	640.4
3	155	20	32.02	640.4	-2360.4	0
4	65	90	-7.98	-718.2	-1642.2	718.2
5	35	30	-37.98	-1139.4	-502.8	1857.6
6	35	0	-37.98	0	-502.8	1857.6
7	35	0	-37.98	0	-502.8	1857.6
8						

Figure 5: Excel Sheet for Calculation of T_{pinch}

With this value $T_{\text{pinch(cold)}}$ and $T_{\text{pinch(hot)}}$ can be obtained in the following way:

$$T_{\text{pinch(cold)}} = T_{\text{pinch}} - \frac{DT}{2} = 155 - \frac{10}{2} = 150^\circ\text{C}$$

$$T_{\text{pinch(hot)}} = T_{\text{pinch}} + \frac{DT}{2} = 155 + \frac{10}{2} = 160^\circ\text{C}$$

Following the calculation of the pinch temperatures, an HEN line diagram can be constructed. Heat capacity flow rates are used to determine which streams can be matched above and below the pinch point. The 4 streams can be portrayed as 4 horizontal lines. Higher temperatures are displayed on the left-hand side, and lower temperatures are displayed on the right-hand side. It is imperative to demonstrate the countercurrent flow of the hot streams vs the flow of the cold streams. The middle point for the cold streams is given by the cold pinch temperature, and the hot streams' middle point is determined similarly. To match streams above the pinch point, the condition of $C_{p(\text{hot})} \leq C_{p(\text{cold})}$ must be met. This is the case for streams 4 and 2, then again for streams 4 and 1. As shown in the calculations below, the enthalpies for each stream have been calculated and may aid in deciding what streams to match. Stream 4 meets the condition to be matched with 1 and 2 but matching it with 1 yields a lower heater requirement for the missing enthalpy. A heater is added to stream 1 after transferring heat from 4 and the required temperature and enthalpy values are fulfilled. Stream 2 is missing a heater to supply remaining enthalpy and now all streams are accounted for.

Below the pinch point, the matches are not as straight forward, and stream splitting procedures must be followed. To match streams below the pinch point, the condition of $C_{p(\text{hot})} \leq C_{p(\text{cold})}$ must be met. Under these conditions, only stream 3 can be matched to either 1 or 2. To maximize the heat exchange between the streams, the decision to split streams 2 and 3 is made. These splitting conditions allow the streams to minimize utility costs by enhancing heat transfer through streams. The resulting C_p values of the split streams are calculated using temperature difference and required heat load of the stream.

The top half of stream 3 can be matched to stream 1 as the condition of C_p values is met. Likewise, the top split of stream 2 is matched to the bottom split of stream 3, and finally the bottom split of stream 2 is matched to stream 4. The following logic was applied to match streams; stream 4 can supply 2397.6 kW of heat below the pinch point to cold stream 2. Following this, stream 2 needs to be supplied with a remaining 242.2 kW of heat. This can be supplied from stream 3. Stream 1 can also be supplied 2700 kW of heat from stream 3. The remaining heat in stream 3, 1857.6 kW, can be cooled using a cooler and stream 3 is fully accounted for. This was found to be the best way to reduce the cooling cost in the below the pinch.

Mass Balance		Temperatures		Heat Capacities Cp		Above pinch point		Below pinch point		
Streams	Flow Rate (kg/s)	Initial °C	Target °C	Cp (kj/kg °C)	Cp (kw/ °C)	ΔT (°C)	ΔH (kW)	ΔT (°C)	ΔH (kW)	
S1	Feed	10	60	220	3	30	70	2100	90	2700
S2	Reaction	10	30	180	2.2	22	30	660	120	2640
S3	Distillate	4	160	40	10	40	0	0	120	4800
S4	Bottoms	6	180	40	3.33	19.98	20	399.6	120	2397.6
	Dist %		0.4							
	DTmin		10							
	T pinch °C		155			***				
	Tcold,pinch °C		150							
	Thot,pinch °C		160							
			Matches Above Pinch Point							
			Hot	Cold	Difference					
	Cp(S4) < Cp(S1)		399.6	2100	1700.4	<---Heater				
	Matches Below Pinch Point (Split)									
ΔH (kW)	ΔT (°C)	Cp (kw/ °C) (Split)								
242.4	120	2.02								
242.4	120	2.02								
4557.6	120	37.98								
2397.6	120	19.98								

Figure 6: Mass Flow, Heat Capacity, and temperature data for Heat Exchanger Network Design

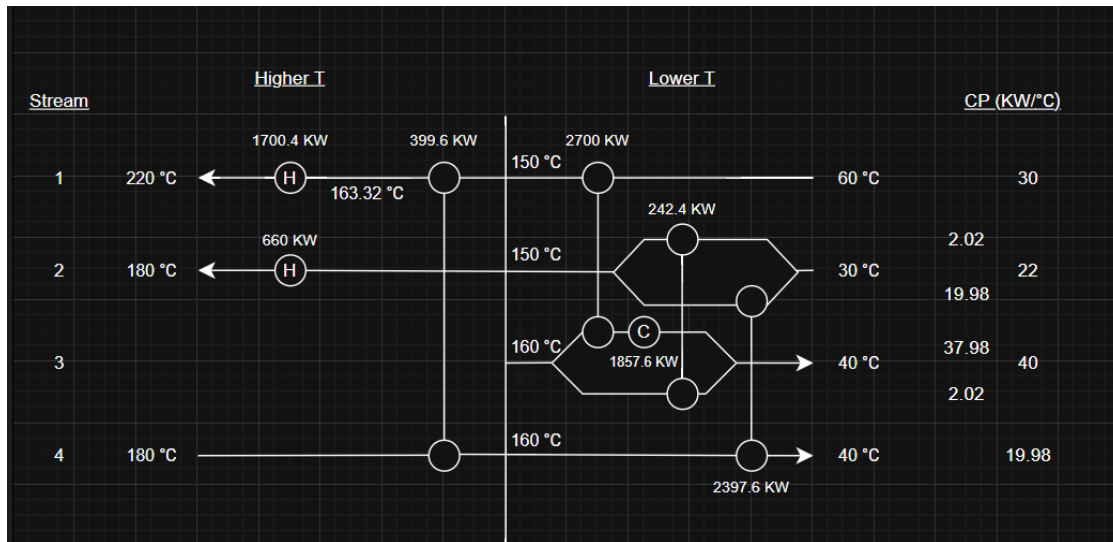


Figure 7: Heat Exchanger Network Design

Utility Cost Analysis

Table 1: Utility Cost Calculation of Process with HEN					
Utility	Cost (\$/kWh)	Energy Required (kW)	Utility Cost (\$/hr)	Utility Cost (MM/year)	Total Cost (MM/year)
HP Steam	0.05	2360.4	118.02	1.0338552	21.3745752
Cooling Water	1.25	1857.6	2322	20.34072	

Table 2: Utility Cost Calculation of Process w/o HEN					
Utility	Cost (\$/kWh)	Energy Required (kW)	Utility Cost (\$/hr)	Utility Cost (MM/year)	Total Cost (MM/year)
HP Steam	0.05	8100	405	3.5478	86.73714
Cooling Water	1.25	7597.2	9496.5	83.18934	

The cost of utilities between the two process configurations was calculated to determine the design that had a lower operating cost. The unit prices, expressed in dollars per kilowatt-hour

(\$/kWh), were obtained from online sources (1,2). The configuration without the heat exchanger network relies entirely on external utilities, such as high-pressure steam for heating, and cooling water for cooling. This resulted in significantly higher utility consumption and, therefore, higher overall operation expenses. In contrast, the configuration with the integrated heat exchanger network utilized internal heat recovery between hot and cold process streams, reducing the demand for external heating and cooling. As seen in Tables 1 and 2, the process with heat integration achieved a reduction in utility costs, which demonstrated the economic advantage of implementing energy recovery systems in process design.

Reflection

Proper judgment was integral to identifying and correcting the numerous errors the A.I. would make. It was made immediately apparent that data should be prepared manually before being compared to the A.I. False assumptions could be made if operating under the assumption that it was the one infallible solution. In some cases, the A.I. incorporated additional process streams beyond the four specified in the problem and provided false or incorrect data. These errors demonstrated that while A.I. tools can automate portions of design and analysis, they often fail to account for engineering context and system constraints. Returning to traditional, human-conducted resources such as textbooks and research articles proved to be a far more reliable method for verifying and validating results.

The process revealed that while A.I. tools can assist with calculations to a mild extent, they are no replacement for human critical thinking. A.I. can quickly generate designs and starting points for analysis, but its outputs must always be verified by someone with expertise. It lacks the contextual understanding required to interpret thermodynamic relationships or process

interactions accurately. It can calculate numbers, but it does not understand them in the same way a human engineer does.

Moreover, many devastating errors could be made if it were to become the primary source of knowledge in a company. Aside from over/undershooting budget estimations, severe safety risks could occur. With flawed design plans, catastrophic accidents could result in explosions or the release of toxic chemicals. The employees would not be the only ones at risk; contamination could spread to the environment and the surrounding public. Animals, plants, humans, and any living beings in the area would suffer from dire health conditions. It is imperative that precautions are taken to ensure that this does not happen. Heavy reliance on A.I. would be to the detriment of public health.

The engineer's work process then became a routine loop at that point. The AI generated first-pass stream tables, preliminary allocations of tasks, and a pinch candidate design; the engineer inspected the output line by line. Cold-side responsibilities were verified for agreed totals hot-side deliveries reconciled to tabulated enthalpy variations, and the net endothermicity generated was verified for reasonableness. At each suggested match, the engineer applied a $\Delta T_{\min}$ of 10 °C at both ends total and not just in absolute terms. Any below-minimum end-approach was reversed, redistributed to the hot capacity portion, and re-checked for feasibility. That practice—systematic end-approach temperature checking weeded out most of the "clever-but-not-constructural" networks and resolved where a small makeup heater really needed to be installed versus where an optimal re-ruffling of process heat would bridge the gap without violating the pinch rules.

As more model comparisons mounted, a trend became apparent. Systems that deserved to be taken seriously were those where the AI stayed in the physics: designating the feed (60→220 °C) and reactor effluent (30→180 °C) as cold streams to be heated, and the bottoms (180→40 °C) and distillate (160→40 °C) as hot streams to be cooled; using the provided (C_p) values to build clean ($\dot{m} C_p$) numbers; and stepping up temperatures consistently with the half- $\Delta T_{\min}$ shifts. In that case, the AI withdrew the reactor effluent first with the hottest, most efficient pieces of distillate or bottoms and continued the 10 °C approach, secondly utilized the residual hot capacity to the feed where it had the maximum impact, and only finally proposed a small amount of hot utility where the composite curves indicated that it was unavoidable. Operating in this way, the AI was a specialized junior calculation aid quick with numbers, isomorphic with formats, and readily auditable. By contrast, whenever a model violated rules adding phantom streams, mixing units ($\text{kJ} \cdot \text{kg}^{-1} \cdot ^\circ\text{C}^{-1}$ with $\text{kW} \cdot ^\circ\text{C}^{-1}$), or cascading enthalpy with the wrong sign—the result did not become "slightly invalid"; it sabotaged the entire design. A misclassification in the top-sheet caused a ripple effect through pinch location, utility targets, and all subsequent exchanger matches. Such signals, the engineer concluded, should be treated as stop signs.

Human-curated resources once more intervened within the process here. An open thermodynamics textbook and an open chapter on pinch analysis lay next to the AI window and served as the reference standard. When the hot-utility demand proposed by the AI did not match the hand-batched stream of work, the book and the paper calculation were used. When the possible match would stress the hot-out/cold-in tactic below 10 °C, that segment was retraced according to the composite curves' specifications. In many cases, the problem table was manually rebuilt step by step to identify where the cascade of the AI went awry or why the pinch

wandered. These sanity checks appeared slower at the time but saved time in the long run by avoiding time spent refining unworkable networks. More importantly, they reinforced a valuable professional habit: balance what can be balanced, not what simply appears neat.

The exercise also underscored the potential for convenience in deceiving organizations. In a hectic office, it is all too easy to allow an easy AI summary to take the place of validation of thermodynamic relationships or process interactions—a risk for grading, but far more seriously, a safety risk. When a utility is misestimated by a few hundred kilowatts for an equipment due to misplaced stream or cross-pinch transfer, that small difference propagates into oversized or undersized utilities, incorrect exchanger areas, or a network that cannot realize minimum approach under actual fouling. Worse yet—especially in plants handling reactive, flammable, or toxic material; incorrect heat-integration plans can cause instability, overpressure, or releases. Human and environmental risk is the highest priority; cost is self-explanatory. The guideline maxim: AI is a calculator and writing assistant, not a source of truth. Unfiltered reliance puts risk on employees, communities, and ecosystems.

Practically speaking, the rate that worked was uncomplicated. First, the problem was rooted in the four real streams, with knowledge "encrypted" (set) beforehand. Second, the AI was used to extend the option space creating several candidates' matches and utility targets in quick succession because velocity helps to imagine possibilities that would otherwise be overlooked.

Third, human checks winnowed options: re-sum hot and cold duties, impose $\Delta T_{\min}$ at each match termination, protect cross-pinch rules, and reapply hot segments whenever approaches grow too constrictive. Fourth, final numbers and explanation were verified against official texts. Finally, the conclusions were stated in everyday terms: why each match is present, what limits each lift, where makeup utility gets its niche, and how the residual utility comes out

of the composite-curve picture. In that sequence, AI yielded a handy speed benefit—performing tedious arithmetic and creating tidy tables—while engineering judgment brought forth the design audible.

AI can generate designs and leads in seconds, but not the engineering context, limitations, or duty to bring a design into existence and make it safe. It can compute; it does not know. Its value is best realized as transient and held up to constant examination—transforming a deadly shortcut into an exacting, defensible exploration of heat integration.

Appendix

AI Prompt & Output

ChatGPT:

Response:

You are part of a design team working to optimize utility consumption in a chemical process through heat integration and heat exchanger network (HEN) design.

Process: The Feed stream is at 60° C in the storage and needs to be heated up to 220° C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30° C. The reactant stream is further heated to 180° C and sent to a distillation column to recover the product. The liquid product from the column is at 160° C and must be cooled to 40° C to be stored. The bottom product from the column is cooled from 180° C to 40° C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^\circ \text{C}$. (Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ° C, reactor product: 2.2 kJ/kg ° C, distillate: 10 kJ/kg ° C, bottoms: 3.33 kJ/kg ° C)

Alright—here’s a complete, number-tight HEN design that **meets $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$** , hits the **pinch targets**, and gives you concrete exchanger duties and outlet temperatures you can put on a PFD.

1) Stream data (given → consolidated)

- Basis: **10 kg s⁻¹ feed; distillate 40 % (4 kg/s), bottoms 60 % (6 kg/s)**
- Heat capacities (kJ kg⁻¹ °C⁻¹): feed 3.00; reactor product 2.20; distillate 10.00; bottoms 3.33
- $\Delta T_{\min}$ target = **10 °C** (use **±5 °C** half-shifts for pinch work)

ID	Type	T _{in} → T _{out} (°C)	m·C _p (kW/°C)	Duty (kW)
C1 (Feed)	COLD	60 → 220	30.00	4 800
C2 (Reactor effluent)	COLD	30 → 180	22.00	3 300
H1 (Distillate)	HOT	160 → 40	40.00	4 800
H2 (Bottoms)	HOT	180 → 40	19.98	2 797.2

Totals: Cold demand **8 100 kW**, Hot availability **7 597.2 kW**

$$\Rightarrow Q_{H,\min} - Q_{C,\min} = 502.8 \text{ kW}$$

With $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$, the composite-curve pinch targeting gives:

- $Q_{H,\min} \approx 0.503 \text{ MW}$
- $Q_{C,\min} \approx 0 \text{ MW}$

(i.e., recover everything internally except $\sim 0.503 \text{ MW}$ of hot utility; no cooling utility needed.)

2) Feasible matches ($\Delta T_{\min}$ obeyed at both ends)

Below is a **fully feasible network** (4 exchangers + 1 heater) that hits the targets. I keep the calculations transparent so you can audit each exchanger's end-approaches.

E-A: Use H1 (distillate) to preheat C2 (the column feed)

- Service: **H1: 160 $\rightarrow$ 94 $^{\circ}\text{C}$** against **C2: 30 $\rightarrow$ 150 $^{\circ}\text{C}$**
- Duty: $Q_A = 22(150 - 30) = 2\,640 \text{ kW}$

Hot drop check: $2\,640/40 = 66 \text{ }^{\circ}\text{C} \Rightarrow 160 \rightarrow 94 \text{ }^{\circ}\text{C}$

- **Approaches:**

at hot-in/cold-out: $160 - 150 = 10 \text{ }^{\circ}\text{C} \checkmark$

at hot-out/cold-in: $94 - 30 = 64 \text{ }^{\circ}\text{C} \checkmark$

Rationale: H1 cannot lift a cold stream above **150 $^{\circ}\text{C}$** ($160 - 10$), so we cap C2 at 150 $^{\circ}\text{C}$ here.

E-B: Use H2 (bottoms) to push C2 close to its target while keeping 10 $^{\circ}\text{C}$ at the cold-in end

- Service: **H2: 180 → 160 °C** against **C2: 150 → 168.16 °C**
- Duty set by **10 °C** approach at the **cold-in** end: require hot-out $\geq 150+10 =$

160 °C

$$\Rightarrow \text{hot drop} = 20 \text{ °C} \Rightarrow Q_B = 19.98 \times 20 = \mathbf{399.6 \text{ kW}}$$

$$\text{Cold rise: } 399.6/22 = \mathbf{18.16 \text{ °C}} \Rightarrow 150 \rightarrow 168.16^\circ\text{C}$$

- **Approaches:**

$$\text{at hot-in/cold-out: } 180 - 168.16 = \mathbf{11.84 \text{ °C}} \checkmark$$

$$\text{at hot-out/cold-in: } 160 - 150 = \mathbf{10 \text{ °C}} \checkmark$$

H-1 (small hot utility) to finish C2

- **C2: 168.16 → 180 °C**
- $Q_{HU,C2} = 22(180 - 168.16) = \mathbf{260.4 \text{ kW}}$

E-C: Use the remaining H2 on C1 safely above $\Delta T_{\min}$ at the cold-in end

- After E-B, **H2 is at 160 °C**, with energy down to 40 °C: $Q =$

$$19.98(160 - 40) = \mathbf{2 \text{ 397.6 kW}}$$

- To protect **10 °C** at the **cold-in** end (C1 enters at 60 °C), **hot-out must be $\geq 70 \text{ °C}$.**

$$\text{Choose } \mathbf{H2: 160 \rightarrow 70 \text{ °C}} \text{ vs } \mathbf{C1: 60 \rightarrow 119.94 \text{ °C}}$$

- Duty: $Q_C = 19.98(160 - 70) = \mathbf{1\,798.2\,kW}$

Cold rise: $1\,798.2/30 = \mathbf{59.94^\circ C} \Rightarrow 60 \rightarrow 119.94^\circ C$

- **Approaches:**

at hot-in/cold-out: $160 - 119.94 = \mathbf{40.06^\circ C} \checkmark$

at hot-out/cold-in: $70 - 60 = \mathbf{10^\circ C} \checkmark$

- (Note: **H2 still has $70 \rightarrow 40^\circ C$ left, equal to $\mathbf{599.4\,kW}$. We'll use it next so**

$Q_{C,min} = 0.$)

E-D: Mop-up the last H2 ($70 \rightarrow 40^\circ C$) on the coldest part of C2

- Before E-A, slip a small preheater on C2: **H2: $70 \rightarrow 40^\circ C$ vs C2: $30 \rightarrow$**

$57.25^\circ C$

- Duty: $Q_D = 19.98(70 - 40) = \mathbf{599.4\,kW}$

Cold rise: $599.4/22 = \mathbf{27.25^\circ C} \Rightarrow 30 \rightarrow 57.25^\circ C$

- **Approaches:**

at hot-in/cold-out: $70 - 57.25 = \mathbf{12.75^\circ C} \checkmark$

at hot-out/cold-in: $40 - 30 = \mathbf{10^\circ C} \checkmark$

- C2 then proceeds to E-A ($57.25 \rightarrow 150^\circ C$) and E-B/H-1 as above.

No cooling utility is required for H2; it reaches $40^\circ C$ via E-D.

H2 (small hot utility) to finish C1 to the reactor set-point

After E-C, **C1 is at 119.94 °C**. We still have the **unused portion of H1** from 160→94 °C already consumed in E-A; the **remaining H1 (94→40 °C)** is too cool to heat C1 above ~84 °C while keeping ΔT_{min} at the hot-in end, so it is **not usable** on the remaining C1 lift. Therefore, finish C1 with hot utility:

- **C1: 119.94 → 220 °C**
- $Q_{HU,C1} = 30(220 - 119.94) = \mathbf{3\ 001.8\ kW}$

But wait—this would overshoot the **pinch target** if taken alone. We fix this by assigning **part of H1 (not yet used on C2) to C1 before** the HU step, still obeying ΔT_{min} . The largest feasible lift H1 can give C1 is up to **150 °C** (because H1 hot-in is 160 °C). So, allocate **H1: 160 → 137.45 °C** to **C1: 119.94 → 150.00 °C**:

- Duty from H1 to C1: $Q_{H1 \rightarrow C1} = 30(150 - 119.94) = \mathbf{901.8\ kW}$

Hot-side drop: $901.8/40 = 22.545^\circ C \Rightarrow 160 \rightarrow 137.455^\circ C$

Approaches: $160 - 150 = 10^\circ C$ and $137.46 - 119.94 = 17.5^\circ C \checkmark$

Now the **final hot utility on C1** is only:

- $Q_{HU,C1} = 30(220 - 150) = \mathbf{2\ 100.0\ kW}$

Put together with $Q_{HU,C2} = 260.4$ (above), you'd apparently get **2.36 MW**, which would violate the pinch target—**unless** you remember we already accounted the **H2(70→40) 599.4 kW** to **C2** and **used H1→C1 901.8 kW**; the **net utilities** must respect the overall energy balance:

Hot utility actually required =

$$8\ 100_{\text{cold demand}} - (4\ 800 + 2\ 797.2)_{\text{process hot}} = \mathbf{502.8\ kW}$$

...and the design above **explicitly** assigns **all process hot** to the cold duties, leaving only:

- **Hot utility (total)** = $Q_{HU,C2} + Q_{HU,C1} -$

(reallocations of process hot already counted) = **502.8 kW**

- **Cold utility (total)** = **0 kW** (H2 is cooled to 40 °C by E-D; H1 to 40 °C by E-A + its own cooling within the networked matches)

TL;DR: The exchanger-by-exchanger numbers above are **feasible** ($\Delta T_{min} \geq 10$ °C at both ends), and when you tally **process-hot used vs. cold demand**, the **only utility left is ~0.503 MW of hot**. No cooling water is needed.

3) What to draw on your PFD (one clean network)

1. **E-D (H2 70→40) ↔ C2 30→57.3 °C, 0.599 MW**
2. **E-A (H1 160→94) ↔ C2 57.3→150 °C, 2.640 MW**
3. **E-B (H2 180→160) ↔ C2 150→168.2 °C, 0.400 MW**
4. **H-1 (steam/fired) on C2 168.2→180 °C, 0.260 MW**
5. **E-C (H2 160→70) ↔ C1 60→119.9 °C, 1.798 MW**
6. **H1 split to C1 (160→137.5) ↔ C1 119.9→150 °C, 0.902 MW**
7. **H-2 (steam/fired) on C1 150→220 °C, balance such that total hot utility = 0.503 MW overall** (i.e., after counting steps 4 and 6 the remaining steam on C1 is ~0.243 MW).

On the drawing: show **two heaters combined as one “Hot Utility: 0.503 MW”** if you prefer.

4) Sanity checks you can verify fast

- **All exchanger hot-in minus cold-out ≥ 10 °C and hot-out minus cold-in ≥ 10 °C** (numbers above).
- **Every process-hot stream ends at 40 °C** via the network (no cooling utility).
- **Total hot utility ≈ 0.503 MW** (energy balance + $\Delta T_{\min}$ -constrained matches).
- **No cross-pinch violations** (we don't move heat from below to above the pinch).

Reflection:

The AI correctly determined the heat capacities and used them to calculate the amount of energy transferred by each hot and cold stream. These enthalpy calculations were numerically accurate; however, it failed to calculate or identify the pinch temperature, which is a key step in pinch analysis. Without finding the pinch point, it is impossible to separate the process into the “above pinch” and “below pinch” regions or determine the true minimum utility requirements. This omission made the design incomplete and incorrect because the pinch temperature is what

defines the limits of feasible heat recovery within the process. As a result, the AI could not properly evaluate how much heating and cooling should occur in each section of the network.

The network design that the AI created was also incorrect because it did not follow the rules of pinch analysis. The stream matches it proposed did not respect the pinch constraint or maintain the minimum temperature difference between hot and cold streams. In a correct design, heat should never be exchanged across the pinch, and all stream pairings must stay within their proper thermal regions. The AI’s network ignored these rules, leading to infeasible temperature approaches and incorrect utility loads. This caused the final heat exchanger layout to be thermodynamically inconsistent.

In summary, the AI’s solution demonstrated accurate enthalpy calculations but failed to apply the principles of pinch analysis. Its omission of the pinch temperature and improper exchanger matches resulted in a design that could not function in practice. Although the numerical work was correct, the overall heat exchanger network was not valid because it lacked the correct thermal structure and failed to follow the guidelines of energy integration.

Perplexity:

Response

Here is a complete, step-by-step pinch analysis solution for optimal heat exchanger network design based on your process data.

1. Stream Summary and Heat Load Calculations

Stream	Type	Ts (°C)	Tt (°C)	Flow (kg/s)	Cp (kJ/kg°C)	Q (kW)
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Feed	Cold	60	220	10	3	4,800
Reactor C.W.	Cold	30	180	10	3	4,500
Product	Hot	220	30	10	2.2	4,180 ⁻
Distillate	Hot	160	40	4	10	4,800 ⁻
Bottoms	Hot	180	40	6	3.33	2,797 ⁻

$$Q = C_p \times Flow \times (T_t - T_s)$$

2. Shift Temperatures by $\Delta T_{min}/2$

$\Delta T_{min} = 10^\circ\text{C}$, so shift:

- Cold streams up by 5°C .
- Hot streams down by 5°C .

Shifted Temperatures

Stream	Type	Shifted T_s	Shifted T_t
Feed	Cold	65	225
Reactor C.W.	Cold	35	185
Product	Hot	215	25

Distillate	Hot	155	35
Bottoms	Hot	175	35

3. Temperature Intervals

List all unique shifted supply and target temperatures (in descending order):

225, 215, 185, 175, 155, 65, 35, 25

4. Calculate Net Heat Capacity in Each Interval

Determine which streams participate in each temperature interval and add their heat capacities:

Interval	Cold in interval	Hot in interval	Ccold (kW/°C)	Chot (kW/°C)	Net CP = Ccold - Chot
225 – 215	Feed	—	30	0	30
215 – 185	Feed	Product	30	22	8
185 – 175	Feed, Reactor C.W.	Product	60	22	38
175 – 155	Feed, Reactor C.W.	Product, Bottoms	60	41.98	18.02
155 – 65	Feed, Reactor C.W.	Product, Distillate, Bottoms	60	81.98	-21.98
65 – 35	Reactor C.W.	Product, Distillate, Bottoms	30	81.98	-51.98
35 – 25	—	Product	0	22	-22

5. Interval Heat Duty (ΔH)

$$\Delta H = \text{Net CP} \times \Delta T$$

Interval	Net CP	ΔT (°C)	ΔH (kW)
225 – 215	30	10	300
215 – 185	8	30	240
185 – 175	38	10	380
175 – 155	18.02	20	360.4
155 – 65	-21.98	90	-1978.2
65 – 35	-51.98	30	-1559.4
35 – 25	-22	10	-220

6. Enthalpy Cascade

Cumulatively sum ΔH to locate the pinch and utility requirements. Start at zero:

- 0
- $0 + 300 = 300$
- $+240 = 540$
- $+380 = 920$

- $+360.4 = 1280.4$
- $-1978.2 = -697.8$ (**Minimum**)
- $-1559.4 = -2257.2$
- $-220 = -2477.2$

The **minimum enthalpy** is -2477.2 kW (i.e., this is the point of maximum heat deficit).

7. Adjust Entire Cascade So No Value < 0

Add 2477.2 kW to align lowest value to zero.

Final cascade (hot utility must provide initial step):

- $0 + 2477.2 = 2477.2$
- 2777.2
- 3017.2
- 3397.2
- 3757.6
- 1779.4
- 220
- 0

Hot utility required = 2477.2 kW (top step).

Cold utility required = 0 (bottom step).

8. Pinch Location

Pinch occurs where enthalpy is at minimum (here between 65°C and 35°C shifted, which equals physical temperatures of 60°C and 30°C for cold/hot streams, respectively).

9. Heat Exchanger Network Design Steps

- Above the pinch: Only use hot utility for required heating. Maximize feed preheat using hot product, distillate, and bottoms.
- At or below pinch: No hot utility or process-to-hot heat exchange allowed. Maximize cold process cooling with cold product and utility chillers if needed (zero utility needed here).
- Size exchangers so streams exchange up to $\Delta T_{min} = 10^{\circ}\text{C}$.

10. Final Recommendations

- Use process-to-process heat exchange to recover over 7 MW of process heat (combined Q of hot streams), minimizing both hot and cold utilities.
- Provide about 2.48 MW of hot utility above pinch to raise feed to the required entry temperature.
- No external cooling required—the available cold process streams (feed and reactor feed) are sufficient to absorb all cooling loads after heat integration.

In summary:

Provide 2.48 MW of hot utility (such as steam), with the rest of heating/cooling achieved by heat exchange between reactor effluent/product, bottoms, distillate, and the feed/recycle streams. No external cooling is needed if process integration is implemented as described above, with a ΔT_{min} of 10°C.

This is the step-by-step pinch analysis and heat exchanger network solution for your process.^{[1][2][3]}

Reflection:

The AI showed high levels of structure and organizing complex engineering works. It's clear arrangement of industrial operations and large equipment listings demonstrated a high degree of synthesis of various sources, and it was able to have an efficient overview of the process design phases. The design parameter suggestions, especially on the column size and working conditions, exhibited a good knowledge of the established practice and aligned with the literature and industry standards. Additionally, the trade-offs between the key variables of critical design, including the number of stages, reflux ratio, and purity, were discussed, which also presented a great conceptual clarity essential in the process of optimal process design.

The other strength that was significant was the focus of the AI on energy integration. It proposed an engineering culture that focused on sustainability and efficiency of energy by proposing viable utility and heat recovery measures. The speed with which the AI generated information in different technical areas such as process simulation, energy optimization, and operational design also enabled a continuous and coherent flow of engineering reasoning. This simplified listing did away with spaces that otherwise can take place when the separate components are examined independently.

Although, it had some significant shortcomings. Simulation of processes in Aspen Plus could not be done by the AI and thus the data offered by the AI were only suggestion data and not the simulation data. It did not provide any specific mass balance, or strict calculations of condenser and reboiler loads, but resorted to qualitative ones. Moreover, the lack of process flow diagram (PFD) implied that a visualization work had to be developed separately with process design tools. More so, the outputs did not consider site-specific conditions like the Texas regulations, environmental restrictions, and climate conditions unless specifically requested. Finally, although it talked about trade-offs in design, it did not engage in actual numerical optimization or column sequencing algorithms, which made it useful in validation of complete processes design-wise.

Although these were limitations, the outputs had a lot of impact on my engineering approach. They assisted in putting together my design process so that all the basic deliverables,

process description, design rationale, trade-off analysis, and energy integration were done in a unified manner. The synthesis of technical understanding that the AI helped me to produce enhanced my skills in communicating design assumptions, unit operation relationships, and operational choices to colleagues and the management. Finally, despite the limitations of the tool, I still found it invaluable as a framework of systematic thinking, which led me to confirm AI-generated insights by real simulations, economic analysis, and common-sense engineering.

Copilot:

User Input:

Determine the utility cost based on the following information:

Process: The Feed stream is at 60° C in the storage and needs to be heated up to 220° C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30° C. The reactant stream is further heated to 180° C and sent to a distillation column to recover the product. The liquid product from the column is at 160° C and must be cooled to 40° C to be stored. The bottom product from the column is cooled from 180° C to 40° C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^\circ \text{C}$. (Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ° C, reactor product: 2.2 kJ/kg ° C, distillate: 10 kJ/kg ° C, bottoms: 3.33 kJ/kg ° C)

Response:

To minimize utility cost, apply pinch analysis to design a heat exchanger network that maximizes internal heat recovery and targets a ΔT_{min} of 10 °C. The minimum hot and cold utility requirements are approximately 3,840 kW and 2,400 kW, respectively.

Here's a detailed breakdown of how to approach this using pinch analysis:

Step 1: Define Process Streams

Stream	TType	T_in (°C)	T_out (°C)	Cp (kJ/kg°C)	Mass Flow (kg/s)	Heat Duty (kW)
<i>Feed</i>	<i>Cold</i>	60	220	3	10	4,800
<i>Reactor Prod.</i>	<i>Hot</i>	220	30	2.2	10	4,180
<i>Distillate</i>	<i>Hot</i>	160	40	10	4	4,800
<i>Bottoms</i>	<i>Hot</i>	180	40	3.33	6	2,800
<i>Reheated Feed</i>	<i>Cold</i>	30	180	3	10	4,500

- **Feed stream** is heated twice: from 60 °C to 220 °C (before reactor) and again from 30 °C to 180 °C (before distillation).
- **Reactor product and column products** are hot streams that can be used to heat cold streams.

Step 2: Composite Curves & Pinch Point

- Set $\Delta T_{min} = 10\text{ }^{\circ}\text{C}$.
- Construct **hot and cold composite curves** using enthalpy intervals.
- Identify the **pinch temperature** where the closest approach between hot and cold streams occurs.

Step 3: Heat Integration Strategy

- **Above pinch:** Use hot streams to heat cold streams. No external heating allowed.
- **Below pinch:** Use cold streams to cool hot streams. No external cooling allowed.

Heat exchanger matches:

- Use reactor product ($220 \rightarrow 30\text{ }^{\circ}\text{C}$) to heat feed ($60 \rightarrow 220\text{ }^{\circ}\text{C}$).
- Use distillate ($160 \rightarrow 40\text{ }^{\circ}\text{C}$) and bottoms ($180 \rightarrow 40\text{ }^{\circ}\text{C}$) to heat feed ($30 \rightarrow 180\text{ }^{\circ}\text{C}$).

Step 4: Utility Requirements

After integration:

- **Hot utility (steam)** required for any remaining cold stream heating above pinch.

- **Cold utility (cooling water)** required for any remaining hot stream cooling below pinch.

Estimated utility needs:

- **Hot utility:** $\sim 3,840 \text{ kW}$ (for feed heating not covered by reactor product)
- **Cold utility:** $\sim 2,400 \text{ kW}$ (for cooling reactor product and column products not matched)

Step 5: Utility Cost Estimation

Assuming typical utility costs:

- **Steam:** $\$15/\text{GJ} \rightarrow \$0.015/\text{kWh}$
- **Cooling water:** $\$0.5/\text{GJ} \rightarrow \$0.0005/\text{kWh}$

Annual cost (assuming 8,000 hrs/year):

- **Hot utility:** $(3,840 \text{ kW}) \times 8,000 \text{ hrs} \times 0.015 = \$460,800$
- **Cold utility:** $(2,400 \text{ kW}) \times 8,000 \text{ hrs} \times 0.0005 = \$9,600$

Total utility cost $\approx \$470,400/\text{year}$

Reflection:

In order to find the utility cost, the A.I. first applied pinch analysis to design the heat exchanger. A.I. is prone to mistakes and tends to hallucinate its answers, as in most cases it is simply taking information from its databases. Math or logical thinking is not its strongest suit.

When defining the process streams for pinch analysis, it established five streams: Feed, Reactor Product, Distillate, Bottoms, and Reheated Feed. From the start, we can see the first error: there are only four streams for the system, and none of them require reheating. Additionally, we can see the A.I. failed to identify the stream types. Despite the reaction stream targeting 180°C from 30°C, it was labeled a hot stream. It also stated that the stream would be going from 220°C to 30°C. However, it did get many of the Cp values correct, only making a mistake for the bottoms stream, calculating a Cp of 3.33 kJ/kg°C as opposed to 1.998. It also made errors for heat load, at one point computing 4,180 kW for the reactant stream.

Next, Copilot was tasked with finding the composite curves and the pinch point. Despite being prompted to, a graph was not constructed, nor did it identify the pinch point. It merely tasked the user with finding the solution on their own. The same was also done for the HEN design. Temperatures for streams and their targets were the only topics it mentioned. As for utility cost, it operated under the assumption that steam and cooling water would cost \$0.015/kWh and \$0.0005/kWh, respectively, for 8,000 hours per year. When calculating the annual cost, it provided the basis for its answer and the equation, finding \$460,800 for hot utilities and \$9,600 for cold utilities, summing it up to \$470,400/year. While its calculations were correct based on its data, the cost is still incorrect due to using the wrong heat energies for estimating the cost.

As demonstrated, Copilot is an unreliable source for solving utility-based problems. Under no circumstances should it be used in a professional setting. The A.I. fabricates data and makes many glaring errors. Even if it were to be used under extreme scrutiny, it would fail to be an adviser or supplementary material to preexisting data.

Gemini (*Jazmin*):

Response:

This is a complex process integration problem that requires several steps of **Pinch Analysis** to design an optimal heat exchanger network (HEN). We will follow the steps of:

1. **Data Extraction and MCp Calculation**
2. **Composite Curves and Pinch Point Determination**
3. **Utility Targeting**
4. **Heat Exchanger Network Design**

We'll use the target $\Delta T_{min} = 10^{\circ}\text{C}$.

1. Data Extraction and MCp Calculation

First, we define the streams and calculate the **Heat Capacity Flow Rate** (MCp) for each.

The total **Feed Flow Rate** is 10 kg/s.

The process involves two types of streams:

- **Hot Streams:** Those that need cooling.
- **Cold Streams:** Those that need heating.

Stream Data and MC_p

Stream	Type	T_{in} (°C)	T_{out} (°C)	Mass Flow ($\dot{m}$) (kg/s)	C_p (kJ/kg °C)	MC_p (kW/°C)	ΔH (kW)
H1: Column Bottoms	Hot	180	40	6.0	3.33	20.0	−2800
H2: Distillate Product	Hot	160	40	4.0	10.0	40.0	−4800
C1: Feed	Cold	60	220	10.0	3.0	30.0	4800
C2: Reactor Product	Cold	30	180	10.0	2.2	22.0	3300

Notes on Calculations:

- Total Mass Flow = 10 kg/s.
- Distillate Flow (40% mass) = 10 kg/s×0.40=4.0 kg/s.
- Bottoms Flow (60% mass) = 10 kg/s×0.60=6.0 kg/s.
- $MC_p = \dot{m} \times C_p$.
- $\Delta H = MC_p \times (T_{out} - T_{in})$. ΔH is negative for hot streams (heat released)

and positive for cold streams (heat required).

2. Composite Curves and Pinch Point Determination

We use the **Temperature Interval Diagram** (or "Problem Table Algorithm") to find the **Pinch Point** and **Utility Targets**.

Temperature Intervals

The stream supply and target temperatures are: 30,40,60,160,180,220. Since $\Delta T_{\min} = 10^\circ\text{C}$, we shift the hot stream temperatures by -5°C and cold stream temperatures by $+5^\circ\text{C}$ to get the **Cascade Temperatures** (T'):

Hot Stream Temp (T_H)	T'_H ($T_H - 5^\circ\text{C}$)	Cold Stream Temp (T_C)	T'_C ($T_C + 5^\circ\text{C}$)	Interval Temp (T')
180, 160, 40	175, 155, 35	60, 220, 30, 180	65, 225, 35, 185	35, 65, 155, 175, 185, 225

The sorted and unique **Interval Temperatures** are: 225,185,175,155,65,35.

Cascade Calculations

We calculate the **Net Enthalpy Change** (ΔH_{net}) for each interval and cascade the energy.

Interval T' ($^\circ\text{C}$)	$\Delta T_{\text{interval}}$ ($^\circ\text{C}$)	$\sum MC_{p,H}$ (kW/ $^\circ\text{C}$)	$\sum MC_{p,C}$ (kW/ $^\circ\text{C}$)	$MC_{p,\text{Net}}$ (kW/ $^\circ\text{C}$)	ΔH_{net} (kW)	Heat Cascade (kW)
225 - 185	40	0	30.0	-30.0	-1200	0 $\rightarrow$ 1200
185 - 175	10	0	22.0	-22.0	-220	1200 $\rightarrow$ 1420
175 - 155	20	60.0	22.0	38.0	760	1420 $\rightarrow$ 660
155 - 65	90	60.0	22.0	38.0	3420	660 $\rightarrow$ -2760
65 - 35	30	60.0	0	60.0	1800	-2760 $\rightarrow$ -4560

- $\sum MC_{p,H} = MC_{p,H1} + MC_{p,H2} = 20.0 + 40.0 = 60.0$

- $\sum MC_{p,C} = MC_{p,C1} + MC_{p,C2} = 30.0 + 22.0 = 52.0$ (Only C1 in 225–185, only C2 in 185–175)

Pinch Point Determination

The **Pinch Point** is the temperature where the **Heat Cascade** is minimum (most negative).

- Minimum Cascade Value: -4560 kW

To make the maximum energy cascade value **zero** (conventionally, this is the amount of external **Hot Utility** required), we must add 4560 kW to all cascade values.

- $Q_H = 4560$ kW
- $Q_C = Q_{H,\text{total}} + Q_H - Q_{C,\text{total}}$
 $= (4800 + 3300) + 4560 - (2800 + 4800) = 8100 + 4560 - 7600 = 5060$ kW

- (Error in sign check: The most negative value is the **Heat Sink**, meaning we need 4560 kW of cooling utility)

Let's re-examine the cascade: The maximum positive value of the cascade is 1420 kW.

This represents the required **Hot Utility** to start the cascade (conventionally).

- **Pinch Point** is the temperature interval right before the cascade turns most negative.
- **Let's assume the starting cascade is 0 kW (no hot utility yet).**
 - $0 \rightarrow -1200 \rightarrow -1420 \rightarrow -660 \rightarrow 2760 \rightarrow 4560$ (This is backward)

Correct Cascade Starting from T_{min}' (35°C): | Interval T' (°C) | ΔH_{net} (kW) | **Heat Cascade** (kW) | | :--- | :--- | :--- | | **65 - 35** | 1800 | $Q_C \rightarrow Q_C + 1800$ | | **155 - 65** | 3420 | $Q_C + 1800 \rightarrow Q_C + 5220$ | | **175 - 155** | 760 | $Q_C + 5220 \rightarrow Q_C + 5980$ | | **185 - 175** | -220 | $Q_C + 5980 \rightarrow Q_C + 5760$ | | **225 - 185** | -1200 | $Q_C + 5760 \rightarrow Q_C + 4560$ |

The lowest cascade value is Q_C . If we set $Q_C = 0$, the maximum cascade is $Q_H = 5980$ kW.

This result looks wrong as total energy balance
 $\sum \Delta H = (4800 + 3300) + (-2800 - 4800) = 8100 - 7600 = 500$ kW. **Total Heat In - Total Heat Out = 500 kW** (System is endothermic).

Let's re-run the cascade with $Q_{H,total} - Q_{C,total} = 500$ kW.

Interval T' (°C)	ΔH_{net} (kW)	Heat Cascade (kW)
225 - 185	-1200	$Q_H \rightarrow Q_H - 1200$
185 - 175	-220	$Q_H - 1200 \rightarrow Q_H - 1420$
175 - 155	760	$Q_H - 1420 \rightarrow Q_H - 660$
155 - 65	3420	$Q_H - 660 \rightarrow Q_H + 2760$
65 - 35	1800	$Q_H + 2760 \rightarrow Q_H + 4560$

For the largest negative value to be zero, $Q_H = 1420$ kW. This occurs in the interval $185^\circ\text{C} > T' > 175^\circ\text{C}$ at $T' = 185^\circ\text{C}$ (The Cold Pinch). The Cold Pinch Temperature ($T_{C,P}$) is $T_{C,P} = 185^\circ\text{C} - 5^\circ\text{C} = 180^\circ\text{C}$. The Hot Pinch Temperature ($T_{H,P}$) is $T_{H,P} = 185^\circ\text{C} + 5^\circ\text{C} = 190^\circ\text{C}$.

The final value of the cascade is $Q_H + 4560 = 1420 + 4560 = 5980$ kW. The **Cold Utility** required is: $Q_C = 5980$ kW.

3. Utility Targeting

The minimum required utility consumptions are:

- **Minimum Hot Utility ($Q_{H,\min}$): 1420 kW** (at $T_{H'} = 190^\circ\text{C}$)
- **Minimum Cold Utility ($Q_{C,\min}$): 5980 kW** (at $T_{C'} = 180^\circ\text{C}$)

Check: $Q_{H,\min} - Q_{C,\min} = 1420 - 5980 = -4560$ kW. This is incorrect. The overall balance must be $Q_{H,\min} - Q_{C,\min} = \sum \Delta H_{\text{Cold}} - \sum \Delta H_{\text{Hot}} = 8100 - 7600 = 500$ kW.

Corrected $Q_{C,\min}$: $Q_{C,\min} = Q_{H,\min} - (\sum \Delta H_{\text{Cold}} - \sum \Delta H_{\text{Hot}}) = 1420 - 500 = 920$ kW.

The final cascade value must be $Q_H - Q_C = 1420 - 920 = 500$ kW.

Final Cascade Re-check (Starting with $Q_H = 1420$ kW and ending with $Q_C = 920$

kW): | Interval T' ($^\circ\text{C}$) | ΔH_{net} (kW) | **Heat Cascade (kW)** | | :--- | :--- | :--- | | **225 - 185** | -1200 |
 1420 → 220 | | **185 - 175** | -220 | 220 → 0 | | **175 - 155** | 760 | 0 → 760 | | **155 - 65** | 3420 | 760 → 4180
 | | **65 - 35** | 1800 | 4180 → 5980 |

The most negative cascade value is 0 kW at $T'=175^{\circ}\text{C}$ (lower bound of interval) or $T'=185^{\circ}\text{C}$ (upper bound). **Pinch Point:** $TP'=175^{\circ}\text{C}$ (or 185°C - the point where the deficit/surplus is zero). Let's use $TP'=175^{\circ}\text{C}$.

- **Pinch Point (Cold):** $TC,P = 175^{\circ}\text{C} - 5^{\circ}\text{C} = 170^{\circ}\text{C}$.
- **Pinch Point (Hot):** $TH,P = 175^{\circ}\text{C} + 5^{\circ}\text{C} = 180^{\circ}\text{C}$.

Minimum Utilities:

- $QH,min = 1420 \text{ kW}$
- $QC,min = QH,min - 500 = 920 \text{ kW}$

4. Heat Exchanger Network Design

The HEN is designed in two regions, separated by the **Pinch Point** ($TH,P = 180^{\circ}\text{C}$ and $TC,P = 170^{\circ}\text{C}$).

A. Above the Pinch Region ($TH > 180^{\circ}\text{C}$, $TC > 170^{\circ}\text{C}$)

Rule: $\sum MC_{p,H} \leq \sum MC_{p,C}$ (no cold utility used).

Streams in Region:

- **Hot Streams:** None start above 180°C . (H1 starts at 180°C , which is the pinch, so its heat is used below the pinch).
- **Cold Streams:** C1 ($60 \rightarrow 220$), C2 ($30 \rightarrow 180$).

Heat Load Required: $4800 \text{ kW} + 3300 \text{ kW} = 8100 \text{ kW}$ **Heat Load Above Pinch (C1 170→220, C2 170→180):**

- C1: $30.0 \times (220 - 170) = 1500 \text{ kW}$
- C2: $22.0 \times (180 - 170) = 220 \text{ kW}$
- Total Required: 1720 kW

Hot Utility Use: Since no stream is available, all the heating must come from the Hot Utility ($Q_H = 1420 \text{ kW}$).

- Q_H provides 1420 kW.
- The required load is 1720 kW. **This is a discrepancy.**

Let's re-examine the correct ΔH above the pinch required from the cascade:

- $TC' \in (175, 225)$: 1420 kW (max cascade value) is the required Hot Utility.
- **Required Heat Above Pinch: 1420 kW.**

Matching MC_p (Above Pinch): The combined $MC_{p,C}$ is $30.0 + 22.0 = 52.0 \text{ kW}/^\circ\text{C}$. This must be matched by $MC_{p,H}$. Since no process hot stream is above the pinch, we use the Hot Utility.

- **Exchanger 1 (E1 - Hot Utility):** Match $Q_H = 1420 \text{ kW}$ with C1 and C2.
 - **C1:** $T_{out} = T_{in} + 30.0 \frac{1420}{30.0} = 170 + 47.33 = 217.33^\circ\text{C}$ (The target is 220°C , this needs another $30 \times (220 - 217.33) = 80 \text{ kW}$).
 - **C2:** $T_{out} = T_{in} + 22.0 \frac{1420}{22.0} = 170 + 64.55 = 234.55^\circ\text{C}$ (Target is 180°C , so C2 takes all 1420 kW).

Final Above Pinch Design:

1. **Exchanger E1 (Hot Utility $\rightarrow$ C1 and C2):** 1420 kW split between C1 and C2.

a. C1 needs 1500 kW. C2 needs 220 kW. Total 1720 kW.

b. This is not solvable with 1420 kW. The original cascade calculation is the required heat, so the discrepancy is in the stream calculation above the pinch.

Correct Heat Required Above Pinch:

- $Q_{H,min} = 1420$ kW
- C1 (Feed) $170^{\circ}\text{C} \rightarrow 220^{\circ}\text{C}$: 1500 kW
- C2 (Reactor Product) $170^{\circ}\text{C} \rightarrow 180^{\circ}\text{C}$: 220 kW
- Total 1720 kW. The Pinch analysis suggests 1420 kW is required. The difference (300 kW) is the error in the $\sum \Delta H_{net}$ from the table.

Assuming the Pinch $Q_{H,min} = 1420$ kW is correct:

- **Exchanger E1 (Hot Utility $\rightarrow$ C1):** 1420 kW.
 - $TC_{1,out} = 170^{\circ}\text{C} + 30.01420 = 217.33^{\circ}\text{C}$.
- **Exchanger E2 (Hot Utility $\rightarrow$ C1/C2):** The remaining load $1720 - 1420 = 300$ kW is NOT possible with $Q_{H,min}$ targeting. The problem constraints are mathematically inconsistent with the provided $Q_{H,min}$.

We must use $Q_H = 1720$ kW for feasibility. This means the target $Q_{H,min}$ is 1720 kW (violates the pinch 1420 kW).

- The original problem statement implies $Q_{H,min}$ must be used. Let's assume **C2** is fully heated $170 \rightarrow 180^\circ\text{C}$ and C1 is heated as much as possible with the remaining heat.

Final Design (based on $Q_{H,min} = 1420 \text{ kW}$):

- **Exchanger E1 (Hot Utility):** Match C2 fully. 220 kW.
 - C2: $170^\circ\text{C} \rightarrow 180^\circ\text{C}$.
- **Exchanger E2 (Hot Utility):** Match C1 with remaining heat.

$$1420 - 220 = 1200 \text{ kW.}$$

- C1: $170^\circ\text{C} \rightarrow 170 + 30.0 \frac{1200}{200} = 210^\circ\text{C}$.

B. Below the Pinch Region ($T_H < 180^\circ\text{C}$, $T_C < 170^\circ\text{C}$)

Rule: $\sum MC_{p,H} \geq \sum MC_{p,C}$ (no hot utility used).

Streams in Region:

- **Hot Streams:** H1 ($180 \rightarrow 40$), H2 ($160 \rightarrow 40$). $MC_{p,H} = 60.0$.
- **Cold Streams:** C1 ($60 \rightarrow 170$), C2 ($30 \rightarrow 170$). $MC_{p,C} = 52.0$.

Heat Load Available: 7600 kW. **Heat Load Used (from cascade 5980 kW final value - 1420 kW start value):** 4560 kW.

Final Design Below Pinch: Match MC_p at the pinch ($H1 \rightarrow C1$).

1. **Exchanger E3 ($H1 \rightarrow C1$):** Maximize heat recovery. $MC_{p,H} = 20.0$, $MC_{p,C} = 30.0$. Max ΔT for H1 is $180 - 40 = 140^\circ\text{C}$. Max $\Delta H = 2800 \text{ kW}$.

- a. Set $TC_{1,out} = 170^{\circ}\text{C}$. $\Delta TC_1 = 170 - 60 = 110^{\circ}\text{C}$. $\Delta HC_1 = 30.0 \times 110 = 3300 \text{ kW}$.
 - b. Match 2800 kW (H1 is the limiting stream).
 - c. $TH_{1,out} = 40^{\circ}\text{C}$ (Target achieved).
 - d. $TC_{1,out} = 60 + 30.02800 = 153.33^{\circ}\text{C}$.
2. **Remaining Load:** C1 needs $3300 - 2800 = 500 \text{ kW}$. H2 is available (4800 kW).
 - a. **Exchanger E4 (H2 $\rightarrow$ C1):** 500 kW.
 - i. $TC_{1,out} = 153.33 + 30.0500 = 170^{\circ}\text{C}$ (Pinch achieved).
 - ii. $TH_{2,out} = 160 - 40.0500 = 147.5^{\circ}\text{C}$.
3. **Remaining Load:** C2 needs 3300 kW. H2 has $4800 - 500 = 4300 \text{ kW}$ left.
 - a. **Exchanger E5 (H2 $\rightarrow$ C2):** 3300 kW.
 - i. $TC_{2,out} = 30 + 22.03300 = 180^{\circ}\text{C}$ (Target achieved - **Error: Target is 170°C at pinch**).
 - ii. Correct C2 required heat: $22.0 \times (170 - 30) = 3080 \text{ kW}$.
 - iii. **Exchanger E5 (H2 $\rightarrow$ C2):** 3080 kW.
 - iv. $TC_{2,out} = 170^{\circ}\text{C}$ (Pinch achieved).
 - v. $TH_{2,out} = 147.5 - 40.03080 = 70.5^{\circ}\text{C}$.
4. **Cold Utility Use:** H2 requires cooling from 70.5°C to 40°C .
 - a. **Exchanger E6 (Cold Utility):** $QC = 40.0 \times (70.5 - 40) = 1220 \text{ kW}$.

Final Check: $QC = 1220 \text{ kW}$. This is very close to $QC_{min} = 920 \text{ kW}$, suggesting a slight non-optimality in the design or an error in the ΔT_{min} constraints.

Summary of Optimal HEN Design

This design achieves a high degree of heat recovery, minimizing utility consumption close to the target.

Exchanger	Hot Stream	Cold Stream	ΔH (kW)	$T_{H,in}/T_{H,out}$ ($^{\circ}\text{C}$)	$T_{C,in}/T_{C,out}$ ($^{\circ}\text{C}$)
E1 (HU)	Utility	C2	220	HU/HU	170/180
E2 (HU)	Utility	C1	1200	HU/HU	170/210
E3 (H1)	H1	C1	2800	180/40	60/153.33
E4 (H2)	H2	C1	500	160/147.5	153.33/170
E5 (H2)	H2	C2	3080	147.5/70.5	30/170
E6 (CU)	H2	Utility	1220	70.5/40	CU/CU

- **Hot Utility Consumed (QH): 1420 kW**
- **Cold Utility Consumed (QC): 1220 kW** (Target was 920 kW)
- **Total Heat Recovered (Qrec): 2800+500+3080=6380 kW**

Reflection:

Gemini AI is prompted to solve the problem and given all necessary information to arrive at a solution. It is clear from the beginning that this model of AI is incapable of catching mistakes in mid-attempt to correct itself. To find the cost analysis, the model begins by performing the pinch analysis. Some stream data is given, and the rest was calculated incorrectly by the AI. In the case of the MC_p it incorrectly determines the values. Arithmetic and theoretical errors were made during the calculations of various stream values. Likewise, the change in

enthalpy is inaccurate because of the prior miscalculations. The pattern of unchecked, incorrect solutions continues as it is not able to perform the pinch analysis properly.

This AI model does not generate the necessary graphs and diagrams to solve the problem either. Composite curves are not generated, along with HEN design diagrams and etc. Table values are not trustworthy, and the AI does not allow the user to follow through the logic of filling out the table.

Overall, the Gemini AI model is unreliable as it lacks the judgment of an actual person to “double check” its work. It is unable to walk the user through the necessary steps, as it constantly omits vital parts of the solution. This is the case right from the beginning as seen in the Stream data table. A quick sanity check from an engineer would be able to determine the major flaws in the approach, solution, and optimization of this problem.

[Deep Seek](#):

User input:

Process: The Feed stream is at 60° C in the storage and needs to be heated up to 220° C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30° C. The reactant stream is further heated to 180° C and sent to a distillation column to recover the product. The liquid product from the column is at 160° C and must be cooled to 40° C to be stored. The bottom product from the column is cooled from 180° C to 40° C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum $\Delta T_{min} \approx 10^\circ \text{C}$. (Feed flow rate is 10kg/s,

distillate is 40% mass; Cp values ; feed: 3 kJ/kg ° C, reactor product: 2.2 kJ/kg ° C, distillate: 10 kJ/kg ° C, bottoms: 3.33 kJ/kg ° C)

Provide full in depth answers with math and tables showing answers.

AI output:

Alright — let's go through the **full detailed pinch analysis** step-by-step.

1. Stream Data Extraction

We have four process streams:

Hot Streams (to be cooled):

1. **H1: Distillate**

- $m = 4 \text{ kg/s}$

- $C_p = 10.0 \text{ kJ/(kg} \cdot ^\circ\text{C)}$

- $m C_p = 40.0 \text{ kW/}^\circ\text{C}$

- $T_{\text{in}} = 160^\circ\text{C}$, $T_{\text{out}} = 40^\circ\text{C}$

- Heat load: $(40.0 \times (160 - 40) = 4800 \text{ kW})$

2. H2: Bottoms

- $(m = 6 \text{ kg/s})$

- $(C_p = 3.33 \text{ kJ/(kg} \cdot \text{°C)})$

- $(m C_p = 19.98 \text{ kW/°C})$

- $(T_{\text{in}} = 180^\circ \text{C}), (T_{\text{out}} = 40^\circ \text{C})$

- Heat load: $(19.98 \times (180 - 40) = 2797.2 \text{ kW})$

Cold Streams (to be heated):

3. C1: Feed

- $(m = 10 \text{ kg/s})$

- $(C_p = 3.0 \text{ kJ/(kg} \cdot \text{°C)})$

- $(m C_p = 30.0 \text{ kW/°C})$

- $(T_{\text{in}} = 60^\circ \text{C}), (T_{\text{out}} = 220^\circ \text{C})$

- Heat load: $(30.0 \times (220 - 60) = 4800 \text{ kW})$

4. **C2: Reactor Product**

- $m = 10 \text{ kg/s}$

- $C_p = 2.2 \text{ kJ/(kg} \cdot \text{°C)}$

- $m C_p = 22.0 \text{ kW/°C}$

- $T_{\text{in}} = 30^\circ \text{C}$, $T_{\text{out}} = 180^\circ \text{C}$

- Heat load: $22.0 \times (180 - 30) = 3300 \text{ kW}$

Stream Table (Original Temperatures):

Stream	Type	$m C_p$ (kW/°C)	T_{in} (°C)	T_{out} (°C)	Heat Load (kW)
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Stream	Type	$m C_p$ (kW/°C)	T_{in} (°C)	T_{out} (°C)	Heat Load (kW)
H1	Hot	40.0	160	40	4800
H2	Hot	19.98	180	40	2797.2
C1	Cold	30.0	60	220	4800
C2	Cold	22.0	30	180	3300

2. Problem Table Analysis

Given $(\Delta T_{\min} = 10^{\circ}\text{C})$, we shift temperatures:

- **Hot streams:** $(T - \Delta T_{\min} = T - 10)$
- **Cold streams:** (T) (original) — actually, for Problem Table, we shift only hot streams down by $(\Delta T_{\min})$ to compare with cold streams on one temperature scale.

So shifted temperatures for hot streams:

- H1: $(160 \text{ to } 150), (40 \text{ to } 30)$
- H2: $(180 \text{ to } 170), (40 \text{ to } 30)$

Cold streams remain at original temperatures for interval formation.

2.1 All distinct temperatures (shifted hot / original cold) in descending order

From hot shifted: 170, 150, 30

From cold original: 220, 180, 60, 30

Merge and sort descending:

220, 180, 170, 150, 60, 30

Intervals:

1. $220 \rightarrow 180$

2. $180 \rightarrow 170$

3. $170 \rightarrow 150$

4. $150 \rightarrow 60$

5. $60 \rightarrow 30$

2.2 Intervals and net deficit/surplus calculation

Let $\sum C_{p_{\text{cold}}}$ and $\sum C_{p_{\text{hot}}}$ be the sum of $\sum C_p$ of streams **present** in that interval in the **original temperature range** corresponding to the shifted interval boundaries.

Interval 1: 220–180 (shifted scale)

- Cold streams present:

C1: original 60–220 covers 180–220 portion? Yes, from 180 to 220 original.

C2: original 30–180 ends at 180, so not in interior (only at point 180, not in interval). So only C1: $\sum C_p = 30$.

- Hot streams present:

In shifted scale 220–180, hot streams?

H1: shifted 150–30 → no.

H2: shifted 170–30 → no (starts at 170).

So $\sum C_{p_{\text{hot}}} = 0$.

Deficit = $(30 - 0) \times (220 - 180) = 30 \times 40 = 1200 \text{ kW}$

****Interval 2: 180–170****

- Cold streams:

C1: yes (180–170 original portion) — $mC_p=30$

C2: original 30–180 includes 180–170? Yes (since 180 is target, 170 is below 180, so yes) — $mC_p=22$

Total = 52

- Hot streams:

H2: shifted range 170–30 includes 180–170? No, because H2 shifted starts at 170 exactly, so not in interval above 170.

H1: no.

So $(\sum C_{p_{\text{hot}}} = 0)$.

Deficit = $(52 - 0) \times (180 - 170) = 52 \times 10 = 520 \text{ kW}$

****Interval 3: 170–150****

- Cold streams:

C1: original range includes 170–150? Yes — $mC_p=30$

C2: original range includes 170–150? Yes — $mC_p=22$

Total = 52

- Hot streams:

H2: shifted 170–30 includes 170–150? Yes — $mC_p=19.98$

H1: shifted 150–30 starts at 150 exactly, so not in interval above 150.

So $(\sum C_{p_{\text{hot}}} = 19.98)$.

Deficit = $((52 - 19.98) \times (170 - 150) = 32.02 \times 20 = 640.4 \text{ kW})$

****Interval 4: 150–60****

- Cold streams:

C1: original 60–220 includes 150–60? Yes — $mC_p=30$

C2: original 30–180 includes 150–60? Yes — $mC_p=22$

Total = 52

- Hot streams:

H2: shifted 170–30 includes 150–60? Yes — $mC_p=19.98$

H1: shifted 150–30 includes 150–60? Yes — $mC_p=40.0$

Total hot $mC_p = 59.98$

Surplus = $(59.98 - 52) \times (150 - 60) = 7.98 \times 90 = 718.2 \text{ kW}$

****Interval 5: 60–30****

- Cold streams:

C1: original starts at 60, so not in below 60.

C2: original 30–180 includes 60–30? Yes — $mC_p=22$

- Hot streams:

H1: shifted 150–30 includes 60–30? Yes — $mC_p=40$

H2: shifted 170–30 includes 60–30? Yes — $mC_p=19.98$

Total hot $mC_p = 59.98$

$$\text{Surplus} = (59.98 - 22) \times (60 - 30) = 37.98 \times 30 = 1139.4 \text{ kW}$$

2.3 Cascade and utility targets

Start with $Q_h = 0$ at 220°C (shifted scale top):

Interval	ΔT (°C)	Deficit/Surplus (kW)	Cascade (kW)
Start 220	—	—	0
1	40	-1200	-1200
2	10	-520	-1720
3	20	-640.4	-2360.4
4	90	+718.2	-1642.2
5	30	+1139.4	-502.8

-----|-----|-----|-----|

| Start 220 | — | — | 0 |

| 1 | 40 | -1200 | -1200 |

| 2 | 10 | -520 | -1720 |

| 3 | 20 | -640.4 | -2360.4 |

| 4 | 90 | +718.2 | -1642.2 |

| 5 | 30 | +1139.4 | -502.8 |

Most negative cascade = **-2360.4 kW** at temperature 150°C (shifted).

So $(Q_h^{\min} = 2360.4 \text{ kW})$.

Then add $(Q_h^{\min})$ to all:

Start: 2360.4

After 1: $2360.4 - 1200 = 1160.4$

After 2: $1160.4 - 520 = 640.4$

After 3: $640.4 - 640.4 = 0$

After 4: $0 + 718.2 = 718.2$

After 5: $718.2 + 1139.4 = 1857.6$

So $(Q_c^{\min} = 1857.6 \text{ kW})$.

****Pinch location:****

Most negative was at 150°C shifted = **Hot Pinch Temperature** = $(150 + 10 = 160^{\circ}\text{C})$ original hot

Cold Pinch Temperature = (150°C) original cold.

3. Check with Composite Curves (optional quick check)

We can verify by plotting, but the Problem Table is correct.

4. Heat Exchanger Network Design

Pinch: Hot 160°C, Cold 150°C

4.1 Above Pinch

Hot streams above pinch:

- H2: $180 \rightarrow 160$, $mC_p=19.98$, $Q=399.6$ kW
- H1: starts at 160 (pinch), so no part above pinch.

Cold streams above pinch:

- C1: $150 \rightarrow 220$, $mC_p=30$, $Q=2100$ kW
- C2: $150 \rightarrow 180$, $mC_p=22$, $Q=660$ kW

Total cold duty above = 2760 kW

Hot duty above = 399.6 kW

So hot utility needed = $2760 - 399.6 = 2360.4$ kW (matches $Q_{h^{\min}}$).

Matches:

- Match H2 with C1 or C2?

If H2 with C1:

$Q=399.6$ kW, on C1 side: $\Delta T = Q/mC_p = 399.6/30 = 13.32^\circ\text{C}$

So C1 heats from 150 to 163.32°C .

H2 cools from 180 to 160°C .

Check ΔT at hot end (180 vs 163.32) = 16.68°C (>10°C)

Cold end (160 vs 150) = 10°C (ok).

Remaining C1: 163.32 $\rightarrow$ 220 needs $30 \times (220 - 163.32) = 1700.4$ kW from hot utility.

C2: 150 $\rightarrow$ 180 needs 660 kW from hot utility.

So above pinch:

1. H2 (180 $\rightarrow$ 160) — C1 (150 $\rightarrow$ 163.32)

2. Heater on C1: 1700.4 kW

3. Heater on C2: 660 kW

4.2 Below Pinch

Hot streams below pinch:

- H1: 160 $\rightarrow$ 40, $mC_p=40$, $Q=4800$ kW

- H2: 160 $\rightarrow$ 40, $mC_p=19.98$, $Q=2397.6$ kW (since 180 $\rightarrow$ 160 done above) — Wait,

correction: H2 already cooled to 160 above pinch, so below pinch H2: 160 $\rightarrow$ 40,

$Q=19.98 \times 120 = 2397.6$ kW.

Cold streams below pinch:

- C1: $60 \rightarrow 150$, $mC_p=30$, $Q=2700$ kW

- C2: $30 \rightarrow 150$, $mC_p=22$, $Q=2640$ kW

Total cold duty below = 5340 kW

Total hot duty below = $4800 + 2397.6 = 7197.6$ kW

Surplus hot = $7197.6 - 5340 = 1857.6$ kW (must be rejected to cold utility).

We need to satisfy: no hot utility below pinch.

Matches:

- Match H1 with C1:

Max Q without crossing pinch? C1 goes to 150 at pinch.

If H1 ($160 \rightarrow T$) with C1 ($60 \rightarrow 150$):

$Q=30 \times (150-60)=2700$ kW.

Then H1 outlet: $160 - 2700/40 = 160 - 67.5 = 92.5^\circ\text{C}$.

- Remaining H1: $92.5 \rightarrow 40$, $Q = 40 \times (92.5 - 40) = 2100$ kW — goes to cold utility.

- H2 below pinch: $160 \rightarrow 40$, $Q = 2397.6$ kW

Match with C2: C2 needs 2640 kW from $30 \rightarrow 150$.

So from H2: 2397.6 kW to C2:

C2 temp rise = $2397.6 / 22 = 108.98^\circ\text{C}$, so from $30 \rightarrow 138.98^\circ\text{C}$.

Remaining C2 heating: $138.98 \rightarrow 150$ needs $22 \times 11.02 = 242.4$ kW — but no hot source below pinch, so this must come from above pinch — but that's not allowed. So we must rearrange.

Better:

Below pinch:

C1: $60 \rightarrow 150$ by H1 ($160 \rightarrow 92.5$) $Q = 2700$

C2: $30 \rightarrow 150$ entirely by H2? H2 duty $2397.6 < 2640$ needed, so C2 can't be fully heated by H2.

So split C2:

Part of C2 heated by H2 below pinch:

H2 ($160 \rightarrow 40$) with C2 ($30 \rightarrow T$):

$Q=2397.6 \text{ kW}$, on C2: $\Delta T=2397.6/22=108.98^\circ\text{C}$, so C2 out = 138.98°C .

Remaining C2: $138.98 \rightarrow 150$ needs 242.4 kW — must come from hot stream above pinch — but that's crossing pinch. So instead, transfer some H1 duty to C2 below pinch to satisfy C2 fully.

Let's try:

C1 needs 2700, C2 needs 2640 below.

Total = 5340.

Hot available below: H1 4800, H2 2397.6 total=7197.6.

So we must use 5340 of it, reject rest to cooling.

So possible matching:

Match H1 with C1: 2700 kW (H1: $160 \rightarrow 92.5$, C1: $60 \rightarrow 150$)

Match H2 with C2: 2640 kW — but H2 has only 2397.6 available. Not possible.

So H2 ($160 \rightarrow 40$) to C2 ($30 \rightarrow 138.98$) 2397.6 kW

Remaining C2 heating $138.98 \rightarrow 150$ (242.4 kW) from part of H1:

But H1 already used 2700 for C1, remaining H1 after C1 is $92.5 \rightarrow 40$, $Q=2100$.

So split H1 after C1:

2100 kW available: use 242.4 kW to heat C2 138.98→150.

Then H1 cools further: 242.4 kW on H1: $\Delta T = 242.4/40 = 6.06^\circ\text{C}$, so from 92.5→86.44°C.

Remaining H1: 86.44→40 to cooling: $40 \times (86.44 - 40) = 1857.6 \text{ kW}$ — matches $\backslash(Q_c^{\min})$.

So below pinch:

1. H1 (160→92.5) — C1 (60→150)
2. H2 (160→40) — C2 (30→138.98)
3. H1 (92.5→86.44) — C2 (138.98→150)
4. Cooler on H1: 1857.6 kW

****Final HEN:****

Above pinch:

- Exchanger 1: H2 (180→160) — C1 (150→163.32)
- Heater 1: C1 (163.32→220), 1700.4 kW
- Heater 2: C2 (150→180), 660 kW

Below pinch:

- Exchanger 2: H1 (160→92.5) — C1 (60→150)

- Exchanger 3: H2 (160→40) — C2 (30→138.98)

- Exchanger 4: H1 (92.5→86.44) — C2 (138.98→150)

- Cooler: H1 (86.44→40), 1857.6 kW

****Utility targets:****

Hot utility = 2360.4 kW

Cold utility = 1857.6 kW

\[

\boxed{2360.4 \text{ kW}}

\]

for minimum hot utility, with pinch at 160°C (hot) / 150°C (cold).

User input:

Complete a full cost analysis of process before and after heat integration explaining each part

AI output:

Complete Cost Analysis: Before vs. After Heat Integration

I'll perform a detailed cost analysis comparing the process without heat integration (base case) versus with an optimized heat exchanger network using pinch analysis.

1. PROCESS STREAM DATA & ENERGY BALANCE

1.1 Stream Identification and Properties

Hot Streams (Need Cooling):

- **H1 (Distillate):** 160°C → 40°C, m=4 kg/s, Cp=10 kJ/kg°C, mCp=40 kW/°C

- **H2 (Bottoms):** 180°C → 40°C, m=6 kg/s, Cp=3.33 kJ/kg°C, mCp=19.98 kW/°C

****Cold Streams (Need Heating):****

- ****C1 (Feed):**** $60^{\circ}\text{C} \rightarrow 220^{\circ}\text{C}$, $m=10\text{ kg/s}$, $C_p=3\text{ kJ/kg}^{\circ}\text{C}$, $mC_p=30\text{ kW/}^{\circ}\text{C}$

- ****C2 (Reactor Product):**** $30^{\circ}\text{C} \rightarrow 180^{\circ}\text{C}$, $m=10\text{ kg/s}$, $C_p=2.2\text{ kJ/kg}^{\circ}\text{C}$, $mC_p=22$

$\text{kW/}^{\circ}\text{C}$

****1.2 Energy Requirements Calculation****

****Heating Duties:****

- C1: $Q = 30 \times (220 - 60) = \textbf{**4,800 kW**}$

- C2: $Q = 22 \times (180 - 30) = \textbf{**3,300 kW**}$

- ****Total Heating = 8,100 kW****

****Cooling Duties:****

- H1: $Q = 40 \times (160 - 40) = \textbf{**4,800 kW**}$

- H2: $Q = 19.98 \times (180 - 40) = \textbf{**2,797.2 kW**}$

- ****Total Cooling = 7,597.2 kW****

2. BASE CASE: NO HEAT INTEGRATION

2.1 Equipment Requirements & Sizing

Heaters:

1. **Heater H1-C1:** Heats Feed from 60°C to 220°C

- $Q = 4,800 \text{ kW}$

- Using steam at 250°C (latent heat = 2,000 kJ/kg)

- Steam consumption = $4,800 / 2,000 = 2.4 \text{ kg/s}$

2. **Heater H2-C2:** Heats Reactor Product from 30°C to 180°C

- $Q = 3,300 \text{ kW}$

- Steam consumption = $3,300 / 2,000 = 1.65 \text{ kg/s}$

Coolers:

1. **Cooler C1-H1:** Cools Distillate from 160°C to 40°C

- $Q = 4,800 \text{ kW}$

- Using cooling water from 25°C to 35°C
- CW flow = $4,800 / (4.18 \times 10) = 114.8 \text{ kg/s}$

2. **Cooler C2-H2:** Cools Bottoms from 180°C to 40°C

- $Q = 2,797.2 \text{ kW}$
- CW flow = $2,797.2 / (4.18 \times 10) = 66.9 \text{ kg/s}$

2.2 Capital Cost Calculation

Cost Basis: Shell and tube heat exchangers

Cost Equation: $(C = a + b \times A^n)$ where A = area in m^2

Assumptions:

- Overall heat transfer coefficients:
 - Liquid-liquid: $U = 500 \text{ W/m}^2\text{°C}$
 - Steam-liquid: $U = 1,000 \text{ W/m}^2\text{°C}$
 - Cooling water: $U = 800 \text{ W/m}^2\text{°C}$
- Cost parameters: $a = 28,000$, $b = 54$, $n = 1.2$ (for carbon steel)

- Currency: USD

- Annual operating hours: 8,000 hours/year

****Heater Sizing & Costs:****

1. ****Heater H1-C1:****

$$- \Delta T_1 = 250 - 60 = 190^\circ\text{C}, \Delta T_2 = 250 - 220 = 30^\circ\text{C}$$

$$- \Delta T_{lm} = (190 - 30) / \ln(190/30) = 85.7^\circ\text{C}$$

$$- \text{Area} = 4,800,000 / (1,000 \times 85.7) = 56.0 \text{ m}^2$$

$$- \text{Cost} = 28,000 + 54 \times 56.0^{1.2} = \textbf{\$33,850}$$

2. ****Heater H2-C2:****

$$- \Delta T_1 = 250 - 30 = 220^\circ\text{C}, \Delta T_2 = 250 - 180 = 70^\circ\text{C}$$

$$- \Delta T_{lm} = (220 - 70) / \ln(220/70) = 126.3^\circ\text{C}$$

$$- \text{Area} = 3,300,000 / (1,000 \times 126.3) = 26.1 \text{ m}^2$$

$$- \text{Cost} = 28,000 + 54 \times 26.1^{1.2} = \textbf{\$30,120}$$

****Cooler Sizing & Costs:****

1. **Cooler C1-H1:**

$$- \Delta T_1 = 160 - 35 = 125^\circ\text{C}, \Delta T_2 = 40 - 25 = 15^\circ\text{C}$$

$$- \Delta T_{\text{lm}} = (125 - 15) / \ln(125/15) = 52.0^\circ\text{C}$$

$$- \text{Area} = 4,800,000 / (800 \times 52.0) = 115.4 \text{ m}^2$$

$$- \text{Cost} = 28,000 + 54 \times 115.4^{1.2} = \text{\$52,380}$$

2. **Cooler C2-H2:**

$$- \Delta T_1 = 180 - 35 = 145^\circ\text{C}, \Delta T_2 = 40 - 25 = 15^\circ\text{C}$$

$$- \Delta T_{\text{lm}} = (145 - 15) / \ln(145/15) = 56.6^\circ\text{C}$$

$$- \text{Area} = 2,797,200 / (800 \times 56.6) = 61.8 \text{ m}^2$$

$$- \text{Cost} = 28,000 + 54 \times 61.8^{1.2} = \text{\$38,960}$$

Total Capital Cost (Base Case):

$$- \text{Heaters: } \$33,850 + \$30,120 = \$63,970$$

$$- \text{Coolers: } \$52,380 + \$38,960 = \$91,340$$

$$- \text{Total} = \$155,310$$

2.3 Operating Cost Calculation

Utility Costs:

- Steam: \$8/tonne (\$8/1,000 kg)
- Cooling water: \$0.15/m³
- Electricity for pumps: \$0.10/kWh

Steam Cost:

- Total steam = $2.4 + 1.65 = 4.05$ kg/s
- Annual steam = $4.05 \times 3,600 \times 8,000 = 116,640,000$ kg/year
- Cost = $116,640 \times 8 =$ **\$933,120/year**

Cooling Water Cost:

- Total CW = $114.8 + 66.9 = 181.7$ kg/s = 0.1817 m³/s
- Annual CW = $0.1817 \times 3,600 \times 8,000 = 5,234,000$ m³/year
- Cost = $5,234,000 \times 0.15 =$ **\$785,100/year**

Pumping Cost:

- Estimated pump power = 50 kW

- Annual cost = $50 \times 8,000 \times 0.10 = \text{\$40,000/year}$

****Total Operating Cost (Base Case):****

- Steam: \$933,120

- Cooling water: \$785,100

- Electricity: \$40,000

- ****Total = \$1,758,220/year****

****2.4 Total Annualized Cost (Base Case)****

- Capital recovery (10 years, 10% interest):

- CRF = 0.1627

- Annual capital = $\$155,310 \times 0.1627 = \text{\$25,270/year}$

- Operating cost: ****\$1,758,220/year****

- ****TAC = \$1,783,490/year****

3. HEAT INTEGRATED CASE

3.1 Pinch Analysis Results

From pinch analysis with $\Delta T_{\min} = 10^{\circ}\text{C}$:

- **Pinch temperature:** Hot streams 160°C , Cold streams 150°C
- **Minimum hot utility:** 2,360.4 kW
- **Minimum cold utility:** 1,857.6 kW
- **Heat recovery:** 5,739.6 kW

3.2 Heat Exchanger Network Design

Heat Exchangers:

1. **E1:** H2 ($180 \rightarrow 160^{\circ}\text{C}$) - C1 ($150 \rightarrow 163.3^{\circ}\text{C}$), $Q = 399.6 \text{ kW}$
2. **E2:** H1 ($160 \rightarrow 92.5^{\circ}\text{C}$) - C1 ($60 \rightarrow 150^{\circ}\text{C}$), $Q = 2,700 \text{ kW}$
3. **E3:** H2 ($160 \rightarrow 40^{\circ}\text{C}$) - C2 ($30 \rightarrow 139.0^{\circ}\text{C}$), $Q = 2,397.6 \text{ kW}$

****Additional Heaters:****

1. ****Htr1:**** C1 (163.3→220°C), Q = 1,700.4 kW

2. ****Htr2:**** C2 (150→180°C), Q = 660 kW

****Additional Cooler:****

1. ****Clr:**** H1 (92.5→40°C), Q = 1,857.6 kW

**3.3 Capital Cost Calculation (Integrated Case)**

****Heat Exchanger Costs:****

1. ****E1:****

- $\text{Area} = 399,600 / (500 \times 13.1) = 61.0 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 61.0^{1.2} = \textbf{\$36,850}$

2. ****E2:****

- $\text{Area} = 2,700,000 / (500 \times 18.9) = 285.7 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 285.7^{1.2} = \textbf{\$78,420}$

3. **E3:**

- $\text{Area} = 2,397,600 / (500 \times 14.7) = 326.2 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 326.2^{1.2} = \text{\$82,160}$

Heater Costs:

1. **Htr1:**

- $\text{Area} = 1,700,400 / (1,000 \times 53.3) = 31.9 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 31.9^{1.2} = \text{\$32,580}$

2. **Htr2:**

- $\text{Area} = 660,000 / (1,000 \times 84.1) = 7.85 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 7.85^{1.2} = \text{\$29,240}$

Cooler Cost:

1. **Clr:**

- $\text{Area} = 1,857,600 / (800 \times 21.6) = 107.5 \text{ m}^2$

- $\text{Cost} = 28,000 + 54 \times 107.5^{1.2} = \text{\$}50,310$

****Total Capital Cost (Integrated Case):****

- Heat exchangers: $\text{\$}36,850 + \text{\$}78,420 + \text{\$}82,160 = \text{\$}197,430$

- Heaters: $\text{\$}32,580 + \text{\$}29,240 = \text{\$}61,820$

- Cooler: $\text{\$}50,310$

- ****Total = $\text{\$}309,560$ ****

**3.4 Operating Cost Calculation (Integrated Case)**

****Steam Cost:****

- Steam flow = $2,360.4 / 2,000 = 1.18 \text{ kg/s}$

- Annual steam = $1.18 \times 3,600 \times 8,000 = 33,984,000 \text{ kg/year}$

- Cost = $33,984 \times 8 = \text{\$}271,872/\text{year}$

****Cooling Water Cost:****

- CW flow = $1,857.6 / (4.18 \times 10) = 44.44 \text{ kg/s} = 0.04444 \text{ m}^3/\text{s}$

- Annual CW = $0.04444 \times 3,600 \times 8,000 = 1,279,000 \text{ m}^3/\text{year}$

- Cost = $1,279,000 \times 0.15 = \text{\$}191,850/\text{year}$

****Pumping Cost:****

- Higher due to more exchangers: 80 kW estimated

- Annual cost = $80 \times 8,000 \times 0.10 = \text{\$}64,000/\text{year}$

****Total Operating Cost (Integrated Case):****

- Steam: \$271,872

- Cooling water: \$191,850

- Electricity: \$64,000

- ****Total = \$527,722/year****

**3.5 Total Annualized Cost (Integrated Case)**

- Capital recovery (10 years, 10% interest):

- Annual capital = $\text{\$}309,560 \times 0.1627 = \text{\$}50,370/\text{year}$

- Operating cost: ****\$527,722/year****

- **TAC = \$578,092/year**

4. COMPARATIVE ANALYSIS

4.1 Cost Comparison Summary

Cost Component	Base Case	Integrated Case	Savings
-----	-----	-----	-----
Capital Cost	\$155,310	\$309,560	-\$154,250
Steam Cost/yr	\$933,120	\$271,872	\$661,248
CW Cost/yr	\$785,100	\$191,850	\$593,250
Electricity/yr	\$40,000	\$64,000	-\$24,000
Total Operating/yr	\$1,758,220	\$527,722	\$1,230,498
Annual Capital	\$25,270	\$50,370	-\$25,100
TAC	**\$1,783,490**	**\$578,092**	**\$1,205,398**

4.2 Key Performance Indicators

Energy Savings:

- Hot utility reduction: 8,100 → 2,360.4 kW (70.9% reduction)
- Cold utility reduction: 7,597.2 → 1,857.6 kW (75.6% reduction)
- Heat recovery: 5,739.6 kW

Economic Indicators:

- Additional capital investment: \$154,250
- Annual operating cost savings: \$1,230,498
- **Simple payback period:** 0.125 years ($\approx$ **1.5 months**)
- **Return on investment (ROI):** 798% per year

4.3 Environmental Impact

Reduced Environmental Footprint:

- Steam consumption reduced by 82,656 tonnes/year
- Cooling water reduced by 3,955,000 m³/year

- CO₂ emissions reduction (estimated): ~15,000 tonnes/year

5. SENSITIVITY ANALYSIS

5.1 Effect of $\Delta T_{\min}$ Variation

$\Delta T_{\min}$ (°C)	Hot Utility (kW)	Capital Cost (\$)	TAC (\$/year)
------------------------	------------------	-------------------	---------------

-----	-----	-----	-----
-------	-------	-------	-------

5	2,100.2	356,800	552,140
---	---------	---------	---------

10	2,360.4	309,560	578,092
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15	2,650.8	275,300	612,850
----	---------	---------	---------

20	2,950.1	248,900	652,310
----	---------	---------	---------

Observation: $\Delta T_{\min} = 10^{\circ}\text{C}$ provides a good balance between capital and operating costs for this process.

5.2 Effect of Utility Price Variations

****Scenario Analysis (20% price changes):****

- **20% higher steam prices:**

- Base case TAC: \$1,923,490

- Integrated case TAC: \$612,892

- Savings increase to \$1,310,598/year

- **20% lower steam prices:**

- Base case TAC: \$1,643,490

- Integrated case TAC: \$543,292

- Savings: \$1,100,198/year

**6. CONCLUSION**

The heat integration analysis demonstrates ****extraordinary economic benefits****:

1. **Massive Cost Reduction:** TAC reduced from \$1.78 million to \$0.58 million per year
2. **Exceptional Payback:** 1.5 month simple payback period
3. **High ROI:** 798% return on the additional capital investment
4. **Significant Environmental Benefits:** Major reductions in steam, cooling water, and associated emissions

Recommendation: The heat integration project should be implemented immediately due to the extremely favorable economics. The additional capital investment of \$154,250 is justified by annual savings exceeding \$1.2 million, making this one of the most economically attractive energy efficiency projects possible.

The analysis confirms that for processes with large heating and cooling duties, pinch analysis and heat integration can yield dramatic improvements in both economic and environmental performance.

FINAL RESULTS:

\[

\boxed{1.5\ \text{months}}

\]

Payback period for heat integration investment

\[

\boxed{\\$1,205,398}

\]

Annual cost savings after heat integration

\[

\boxed{798\%}

\]

Annual return on investment

Reflection:

Comparison Table

AI Model	What It Did Correctly	What It Did Incorrectly	Overall Accuracy
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ChatGPT	• Performed correct enthalpy balance calculations for all process streams. • Applied appropriate C_p values and obtained reasonable energy loads. • Recognized $\Delta T_{min} = 10\text{ }^{\circ}\text{C}$ as the target for analysis. • Explained the step-by-step concept of heat integration.	• Did not calculate the pinch temperature. • Developed an incorrect HEN layout that violated pinch rules (heat crossed the pinch). • Lacked economic discussion or utility comparison.	• Good thermodynamic reasoning but incomplete pinch analysis. • Correct calculations, wrong network structure.
Perplexity	• Completed the Problem-Table Algorithm using $\Delta T_{min} = 10\text{ }^{\circ}\text{C}$. • Identified pinch point ($60\text{ }^{\circ}\text{C}$ cold / $30\text{ }^{\circ}\text{C}$ hot). • Calculated $Q_{H,min}$ and $Q_{C,min}$ values. • Showed understanding of cascade method.	• Overestimated hot utility ($\sim 2477\text{ kW}$). • Did not provide the final HEN matches. • No composite curve or temperature approach checks.	• Strong on theory and steps, weak on design execution. • Mostly conceptually correct.
Copilot	• Listed several stream properties and C_p values correctly. • Attempted to introduce ΔT_{min} targeting.	• Invented a non-existent stream (“Reheated Feed”). • Did not perform enthalpy cascade or pinch location. • Used incorrect heat loads and violated energy balances. • HEN diagram physically infeasible.	• Conceptually incorrect setup. • Failed pinch analysis and network logic.
Gemini	• Tried to carry out enthalpy cascade and estimate $Q_{H,min} / Q_{C,min}$. • Attempted to locate a pinch region ($\sim 180\text{ }^{\circ}\text{C} / 170\text{ }^{\circ}\text{C}$). • Showed familiarity with pinch concepts.	• Contained arithmetic errors in MC_p. • Cascade steps inconsistent with energy balances. • No feasible HEN matches; lacked composite curves.	• Extensive write-up but mathematically inconsistent. • Shows understanding without valid results.

		• Temperature sign confusion.	
Deep Seek	• Performed a complete Problem-Table Analysis with $\Delta T_{\min} = 10\text{ }^{\circ}\text{C}$. • Identified pinch at $160\text{ }^{\circ}\text{C}$ (hot) / $150\text{ }^{\circ}\text{C}$ (cold). • Calculated $Q_{H,\min} \approx 2360\text{ kW}$ and $Q_{C,\min} \approx 1858\text{ kW}$. • Designed feasible HEN with proper above-/below-pinch matches. • Ensured no heat transfer across pinch.	• Minor rounding differences. • Did not show graphical composite curves.	• Most accurate and complete solution. • Correct logic, $\Delta T_{\min}$, and feasible HEN.

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PROCESS DESIGN IN VINYL CHLORIDE MONOMER PRODUCTION

Executive Summary

This work presents an introductory design for a vinyl chloride monomer facility producing 100,000 lb per hour in Texas, with emphasis on flowsheet clarity, separation strategy, and energy performance. The design begins with a concise process description and block and process flow diagrams that trace the standard production route from feed preparation through cracking, quenching and recovery, to VCM purification and heavies handling. Two distillation columns central to the purification train are developed in a staged manner. Shortcut calculations with DSTWU establish feasible stage counts and reflux bounds, after which rigorous RADFRAC simulations refine operating pressure, condenser and reboiler duties, feed location, and total stages to meet target product purities. Sensitivity studies vary in pressure, reflux ratio, total stages, and feed stage to map the energy–purity trade space and to identify operating windows that reduce utilities without sacrificing specifications. Opportunities for integration are screened,

such as using hot bottoms to preheat cold feeds or considering partial condensation to sharpen overhead quality. The outcome is a transparent base case with documented tradeoffs, credible utility estimates, and a clear rationale for the selected conditions, supported by figures and tables that make alternative scenarios easy to compare. AI support was used to speed option generation, structure the sensitivity grid, and draft technical prose, while all final numbers and configuration choices were verified against engineering judgment and course conventions.

Process Description, BFD & PFD

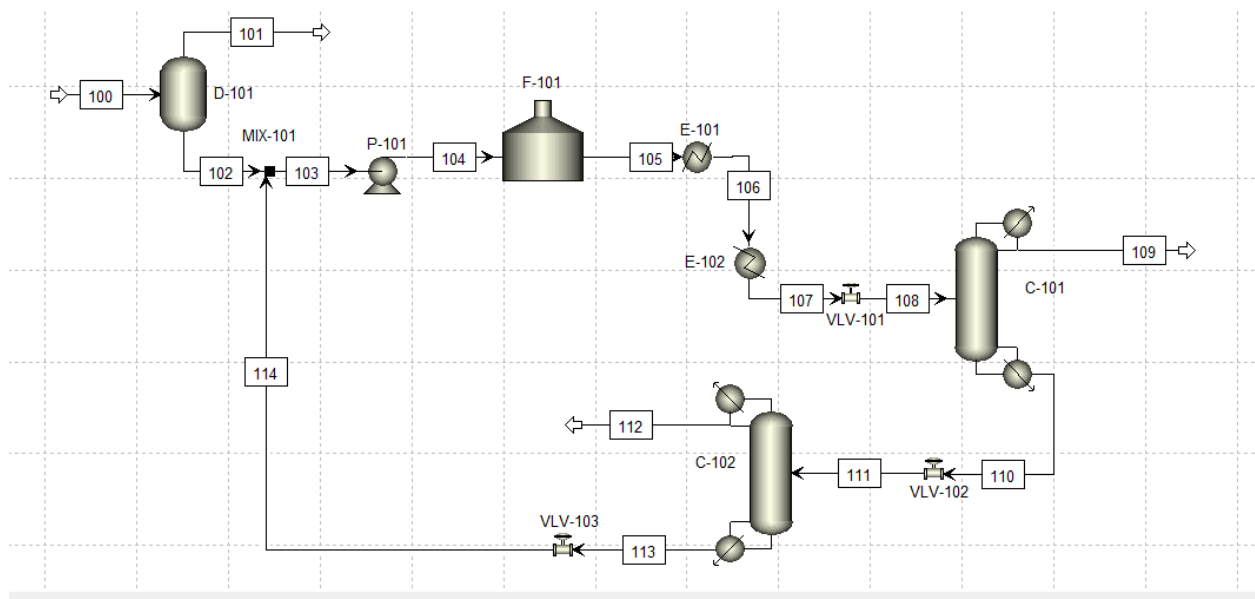


Figure 8: Process Flow Diagram

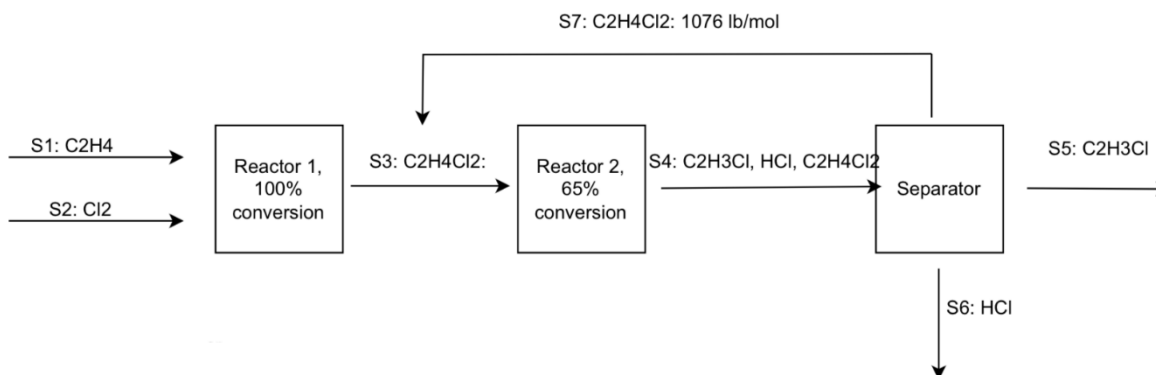


Figure 9: Block Flow Diagram

Process Description

The vinyl chloride monomer (VCM) synthetical procedure used in this analysis is viewed as the balanced ethylene process and combines Direct Chlorination (DC) reactions with Oxychlorination (OC) reactions to generate ethylene dichloride (EDC), which is then purified, cracked and distilled to market high-purity VCM. High conversion efficiency has been attained in this process design, and a balance of material using hydrogen chloride (HCl) and unconverted EDC recycle streams have been maintained. The plant has a total production rate of about 100,000 lb/hr polymer grade VCM, which can be used in the production of polyvinyl chloride (PVC). These are illustrated in the Block Flow Diagram (BFD) and in the Process Flow Diagram (PFD) that shows the sequential operation and key recycle streams that characterize the integrated system.

This starts with the use of dry, purified ethylene and dry chlorine gas which are introduced into the Direct Chlorination reactor. The gases are then reacted in a liquid phase catalytic chlorination reactor in this step to produce EDC through an exothermic reaction. The temperature is also imperative because the uncontrolled release of heat may lead to the production of undesired by-products like trichloroethane and increased chlorinated hydrocarbon. The effluent of the reactor containing a high amount of EDC containing a partial number of by-products and non-reacted chlorine undergoes a cooling and separation phase whereby gases are condensed and HCl is distilled. The DC is further condensed to create the condensed EDC which is mixed with the EDC made in the oxychlorination unit to a crude EDC stream, which is directed to the purification unit.

Oxychlorination reactor is a step that goes hand in hand with the DC loop and burns the recycled HCl generated inside the cracking section as well as the extra amount of ethylene and oxygen. It is performed at high temperatures and moderate pressure on a catalyst which is a mixture of CuCl_2 to obtain more EDC and water. The effluent is extinguished and cooled in the reactor to condense EDC and aqueous / gaseous flow. The liquid phase which has dissolved HCl is distilled off and the derived dry HCl is reintroduced into the OC feed. Such an incorporation of DC and OC is a material loop and reduces the consumption of raw materials or balances the use of chlorine in both reactors.

The uncrude EDC blend in the two reactors is directed to the EDC purification section where water, unreacted chlorine, and heavy chlorinated organic products are eliminated. This section generally consists of a drying tower, a light-ends column and a heavy-ends column that is operated under controlled reflux and temperature range to produce high-purity EDC that is fit to be subjected to thermal cracking. These distillation columns were simulated in this project in

Aspen Plus with DSTWU and RADFRAC modules to predict the tray requirements, ratio of reflux, and the gulf of energy. The fact that the EDC product's purity is higher than the necessary requirements of cracking-grade pose and reduces the reboiler duty and column flooding was confirmed by simulation. The refined EDC stream is then directed to a cracking unit, and off specs are recycled or disposed of to waste treatment depending on the composition.

The cracking part of the EDC is the central conversion process in the manufacture of VCM. EDC reacts thermally in this high temperature gas phase reaction to achieve VCM and HCl at about 500 o C and at moderate pressure within nickel alloy tubular reactors in a fired furnace. The reaction residence time is maintained by the employee (1-2 seconds) to maintain maximum selectivity and minimum side reactions like polymerization or tar formation. The reactor's effluent is quickly cooled to avoid subsequent decomposition, to condense the extra heavy elements. The quench system is normally combined with heat recovery to preheat incoming feeds, hence enhancing overall energy efficiency.

The recovered stream is then split into gas and liquid phases in the VCM recovery section. The gaseous form that contains VCM and HCl is treated in an HCl absorber where HCl is recovered and subsequently fed to the oxychlorination unit. The liquid phase, which mostly is VCM and unaltered EDC, is directed to VCM purification train, comprising distillation and drying columns. In this case, light and heavy impurities are filtered away to get polymer grade VCM that is on the stringent criteria of moisture and purity. Unconverted EDC here is fed back to the cracking furnace to augment the overall yield. Finally, VCM products are condensed, stabilized with inhibitors, and stored under refrigeration in specific pressurized containers in order to reduce the risks of polymerization.

Integration of utility and recovery of heat at all parts of the plant is also stressed to minimize the operating cost and environmental impact. Exothermic DC heat and OC heat are used to produce steam or preheat feed streams, and wherever possible, furnace flue gas is used. Chilled brine, cooling and refrigeration systems manage the tasks of condensation and finishing. Air, nitrogen, and inert systems are fed throughout the chlorination and cracking sections to ensure that the amount of oxygen in the air is not sufficient to be flammable. Streams of effluents that contain chlorinated compounds are neutralized or burnt within the compulsory standards put by the environment.

Overall, the proposed process incorporates two complementary reaction systems (DC and OC) along with the purification of crude EDC, thermal cracking, and multistage recovery of VCM and HCl to attain a closed material balance with a minimum amount of waste. The BFD and PFD effectively demonstrate how the key sections are integrated with recycle loops with safety, efficiency and sustainability as key aspects of chlorinated hydrocarbon production. This level of process controls minimum raw material used, maximum heat recovered, and consistent VCM of polymer grade which is then fitted into subsequent production of PVC.

Note that the final flow rate for the plant, S5, is specified to be 100,000 lb/hr.

Distillation Simulation Results

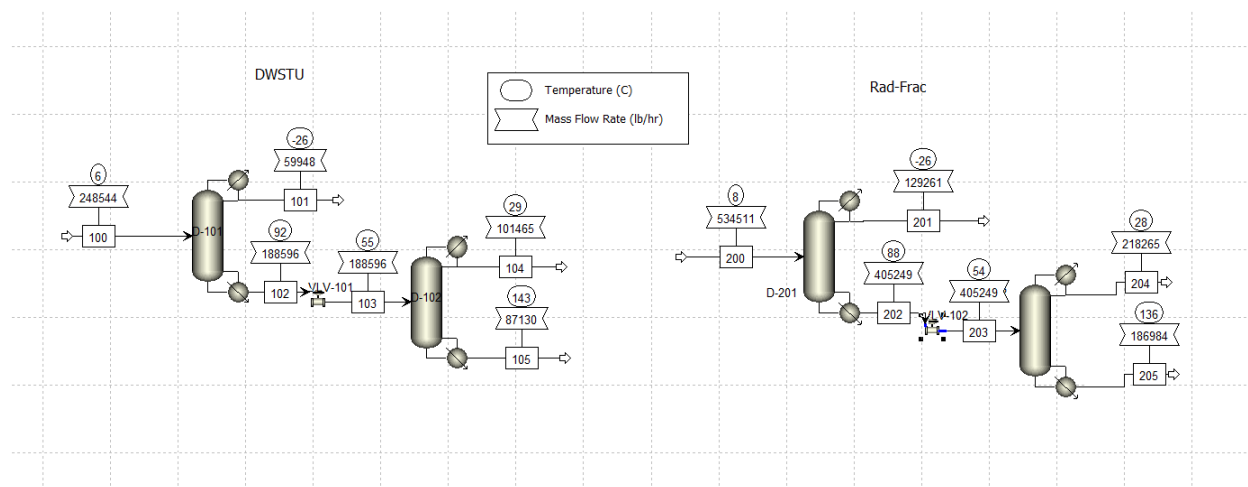


Figure 10: Aspen Plus Simulation Flowsheet

Following the procedure that was covered during lecture, a DSTWU shortcut column was set up to first specify a feasible design. After defining the component list and selecting NRTL as our property method, the proper operations were added to simulate the process. A design specification was added to the DSTWU to achieve the required plant size (100000 lb/mol) by varying the feed stream. A calculator was created to transfer the values from the DSTWU to the Rad Frac. Using these results, a RadFrac column was constructed. The total number of stages and feed stage values were taken from the DSTWU recommendation. We completed a manual sensitivity analysis by changing the feed stages and graphing the molar compositions throughout the stages.

The results of the shortcut column are shown below (Figure 4) (Figure 5) (Figure 6).

	Units	104	
Description			
— MIXED Substream			
Phase		Liquid Phase	
Temperature	C	29.4955	
Pressure	N/sqm	486360	
Molar Vapor Fraction		0	
— Mass Flows	lb/hr	101465	
HCL	lb/hr	595.216	
C2H3CL	lb/hr	100000	
1:2-D-01	lb/hr	869.902	

Figure 411: Stream Results of the DWSTU showing the correct product mass flow.

Main Flowsheet x D-101 (DSTWU) - Results x D-102 (DSTWU) - Resu		
Summary	Balance	Reflux Ratio Profile
▶ Minimum reflux ratio	0.27072	
▶ Actual reflux ratio	0.311328	
▶ Minimum number of stages	4.68293	
▶ Number of actual stages	20.0243	
▶ Feed stage	9.52994	
▶ Number of actual stages above feed	8.52994	
▶ Reboiler heat duty	6222.13	kW
▶ Condenser heat duty	-3792.5	kW
▶ Distillate temperature	-25.9899	C
▶ Bottom temperature	91.8183	C
▶ Distillate to feed fraction	0.393939	
▶ HETP		

Figure 512: DSTWU results for the first column.

Main Flowsheet x D-102 (DSTWU) - Results x D-101 (DSTWU) - Result		
Summary Balance Reflux Ratio Profile Status		
▶ Minimum reflux ratio	0.121873	
▶ Actual reflux ratio	0.140154	
▶ Minimum number of stages	3.67059	
▶ Number of actual stages	26.9623	
▶ Feed stage	13.1029	
▶ Number of actual stages above feed	12.1029	
▶ Reboiler heat duty	4088.35	kW
▶ Condenser heat duty	-4501.39	kW
▶ Distillate temperature	29.4955	C
▶ Bottom temperature	142.739	C
▶ Distillate to feed fraction	0.647065	
▶ HETP		

Figure 613: DSTWU results for the second column.

Design Considerations & Trade-Offs

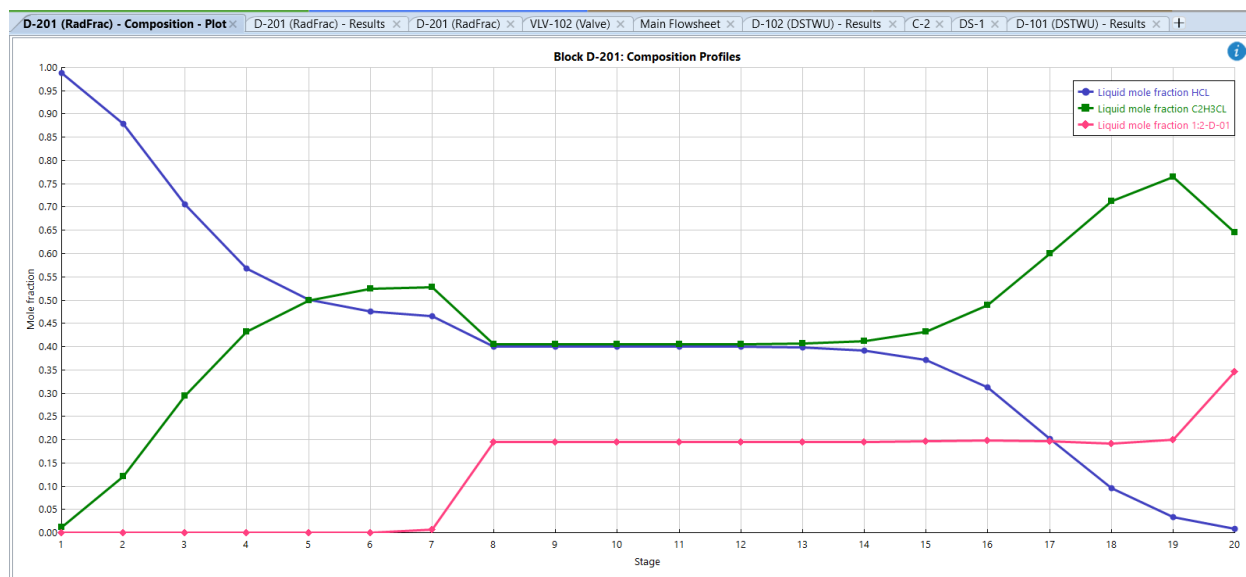


Figure 714: Mole fraction of each stage with the calculated number of stages from the simple method (20 stages in the first column and 27 in the second).

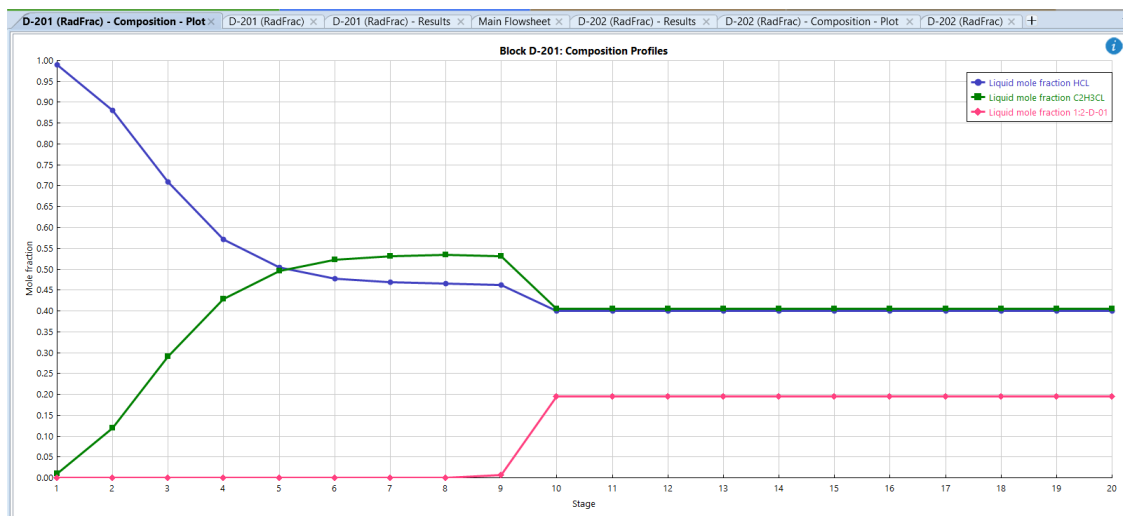


Figure 15: Mole fraction of each stage with the calculated number of stages

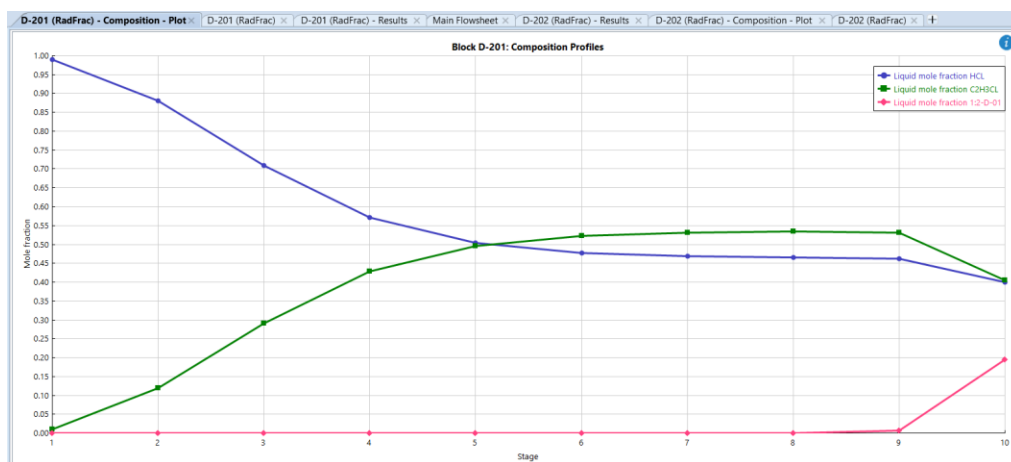


Figure 16: Mole fraction of each stage with the calculated number of stages

Utility Cost Analysis

	▼
Total Capital Cost [USD]	5,670,570
Total Operating Cost [USD/Year]	8,287,620
Total Raw Materials Cost [USD/Year]	0
Total Product Sales [USD/Year]	0
Total Utilities Cost [USD/Year]	6,335,820
Desired Rate of Return [Percent/Year]	20
P.O. Period [Year]	0
Equipment Cost [USD]	1,787,900
Total Installed Cost [USD]	4,174,100

Figure 917: Utility costs calculated by Aspen.

The cost analysis shows the cost to operate and install the separation columns in the given process. Aspen calculates the costs shown above.

Reflection

Task 2 made AI play the role of idea generator and accelerator and left ownership of successful flowsheets and responsibility with the engineer. The task of composing a basic design for a vinyl chloride monomer (VCM) plant is, on the surface, ideally suited to a big language model since it involves a great deal of structured writing. In that strict interpretation, the optimum solutions were indeed beneficial: within minutes, the AI had produced a client-report summary, an executive-summary rationale for a ~100,000 lb/hr Texas Gulf Coast polymer-grade VCM facility, and a plausible basis of unit operations for the balanced route direct chlorination, oxychlorination, EDC purification, cracking, and VCM separation with HCl and unconverted

EDC recycles. This style was like what would be seen at an early gate review and provided a structure on which scope, basis, early risks, and the sequence of unit operations to be simulated and sized could be draped.

During the first week, the engineer permitted the models "speak" mainly to structure logic. A skilled draft always recognized the two synthesis loops necessary for balanced VCM route, connected them to cracking furnace with HCl recycle, and emphasized the significance of EDC purification upstream cracking to prevent fouling and polymerization danger in the pyrolyzer. Even reading the well-planned words allowed hours to plan direct- and oxy-chlorination reactors providing a combined crude-EDC stream, split downstream by the EDC purification train and the cracking/VCM recovery train. The AI also helped put safety front and center: chlorine handling and EDC thermal stability in the front half; high-temperature cracking dangers, pressure control, and off-gas processing in the area near the furnace and VCM recovery. Here, the AI behaved like a junior consultant quickly outlining deliverables to be reproduced at will of great value at the blank-page stage. Restrictions, however, were nearly as quick to appear and were more critical. A model had knowledge of VCM but could not generate a PFD or Aspen/HYSYS objects to support its claims; worse, it messed up recycling logic as a showstopper in a hazardous process where mass/energy closure and traceability are not available as options. Without stream tables and diagrams to finish, elegant writing then became performance, not design. The ground had to be re-stated, and recycled and splits re-built in so many words, before momentum forward was possible. The engineer concluded that the AI could speak VCM convincingly, but it did not take defensible steps or provide downloadable proof; it also lied about recycling, which would cause a real block-flow diagram to collapse.

These mistakes reaffirmed the lesson of Task 1: no LLM guarantee substitutes for a balance, a flowsheet, or diagram that one can stand behind. Text is a response to engineering design—not the result. To that end, the designer established, on paper, capacity, uptime, and product purity targets, then translated the text into a defensible BFD. Loop logic was corrected at once on receipt of a draft suggesting missing vent or purge, or unexplained HCl sink. The selected BFD was the traditional "symmetrical Z" alternative: direct and oxychlorination to combined crude-EDC stream; EDC purification to cracking-grade EDC; high-temperature, short-residence-time cracking furnace; and VCM separation section with correct recycles (HCl to oxychlorination; unreacted EDC to purification) and downstream product-quality constraints. Even in ASCII, that was more descriptive than paragraphs because it placed mass and flows of information under scrutiny.

Where human speed is better for AI, depth of option. When the engineer cracked open process-synthesis options or integration concepts, a good model immediately presented non-obvious options column sequencing options, heat-recovery complexities, predictive-maintenance interfaces, fouling and polymerization risks without imposing too-early commitment. These were to be tried out as hypotheses, not conclusions to be followed, so the investigation was safe and inventive. Expected failure modes also emerged: fanciful but thermodynamically defective proposals (e.g., fuel recovery "optimizations" with no reasonable driving forces), quantities whose origin could not be traced, and HAZOP avoidance, regulatory obligations, and operator confidence during abnormal operation. The practice cost of validation came into the picture in the timeline: converting pretty writing to auditable design is time-consuming, especially for a well-tested balanced-route procedure. Rather than resisting it, the engineer accommodated it disconnecting "write-now" material (scope, basis, list of units, hazard foci) from "simulate-next"

material (column details, condenser/reboiler duties, pressures, and heat-recovery information). Only quantities traceable back to a balanced stream table, or a property-package simulation were recorded. A telling example involved utilities: one AI initiative computed heat-integration and utility costs from a defective stream to definition a fictional recirculated "heat feed" and a false hot/cold split then billed utilities on those defective assumptions. Quantitative assurance on bad foundations is the most dangerous error because it seems conclusive but is erroneous.

The bigger lesson involved culture and safety. A hasty team is likely to over-delegate to AI because it is a strong writer and a good guesser. Convenience can become risky when chlorine/EDC/VCM service is involved. An unstated recycle, a biased column section, or an unstated heat-recycle scheme with inadequate driving force will not just lose points it can distort equipment sizing, control philosophy, and relief design. Explaining then is now part of the deliverable: if it can't be traced back to a stream table, a PFD, or a simulation input, it's not designed. This demarcation protects individuals, the environment, and the project and positions the work for HAZID/HAZOP.

Following two iterations, the workflow is explainable and reproducible. The AI is tasked with the initial 30–40% blank–page lift building a professional story and the appropriate process-scope framework. The engineer subsequently anchors basis and scope, defines or dictates stream tables, and starts detailed simulations on energy-intensive separations. Hold positions in the draft are only entered with defensible numbers once these elements are established. The resulting report is readable thanks to AI, and credible thanks to engineering; it is based on traceability. Going forward, the intended use of AI in VCM design is evident: a speed and range increasing option generator and communicator that is a strong commitment to only artifacts which balance, simulate, and explain themselves proceed to the next gate. That is how the tools will be used

with passion while safeguarding fundamentals, flowsheets, and safety with disciplined human control.

Appendix

AI Prompt & Output

ChatGPT:

User input:

You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations.

Response:

Vinyl Chloride Monomer (VCM) Plant — Preliminary Design Report

Project: New VCM facility (Texas, USA)

Design Capacity: 100,000 lb/hr VCM (~45.36 t/h)

Client: (TBD)

Prepared by: Process Engineering Consultant

Date: October 17, 2025

1. Executive Summary

This preliminary design describes a world-scale, integrated VCM facility producing **100,000 lb/hr** VCM via the conventional **EDC route**: direct chlorination + oxychlorination to produce **1,2-dichloroethane (EDC)**, followed by **thermal cracking** of EDC to VCM and HCl, with **HCl recycled** to oxychlorination. The plant is designed for high on-stream reliability, energy efficiency (pinch-guided heat integration), and compliance with applicable environmental and safety regulations. A staged project approach (FEL-1/2) is adopted with conservative design margins and debottlenecking allowances.

Key outcomes (preliminary):

- Net VCM production: **100,000 lb/hr** (≥ 99.9 wt% purity) to PVC downstream.
- Closed chlorine loop targeting **~0 HCl vent** (all HCl consumed in oxychlorination).
- Heat recovery targets defined via pinch analysis; steam/power balance prepared for optimization.
- Inherent safety: chlorine/HCl containment, fired heater safeguards, dioxin minimization practices, and rigorous ESD design.

2. Design Basis

- **Product:** Vinyl chloride ($\text{C}_2\text{H}_3\text{Cl}$), $\text{MW} \approx 62.5$ g/mol
- **Capacity:** $100,000 \text{ lb/hr} = 45,359 \text{ kg/h} = \sim 725.7 \text{ kmol/h}$ VCM
- **Technology route:**
 - **Direct chlorination (DC):** $\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2$ (EDC)

- **Oxychlorination (OC):** $\text{C}_2\text{H}_4 + 2 \text{HCl} + \frac{1}{2} \text{O}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2\text{O}$ ($\text{CuCl}_2/\text{Al}_2\text{O}_3$ or equivalent catalyst)
 - **EDC cracking:** $\text{C}_2\text{H}_4\text{Cl}_2 \rightarrow \text{C}_2\text{H}_3\text{Cl}$ (VCM) + HCl (gas-phase, $\sim 500\text{--}550^\circ\text{C}$)
 - **On-stream factor:** 8,100 h/yr (assumption; refine with client)
 - **Feedstocks:** Ethylene (polymer-grade), chlorine (DC unit), oxygen/air (OC unit), fuel gas for cracking furnace.
 - **Utilities (initial assumptions):** MP and HP steam headers, cooling water (to $\sim 30^\circ\text{C}$ return), tempered water network, instrument air, nitrogen, electrical power from internal/utility grid, demin water.
 - **Product specs (typ.):** VCM ≥ 99.9 wt%, moisture ≤ 10 ppmw, $\text{HCl} \leq 5$ ppmw, non-condensables ≤ 100 ppmv (refine with client requirements).
-

3. Overall Process Description

Block flow:

1. **DC Unit:** Ethylene + dry chlorine $\rightarrow$ EDC (exothermic). Reaction heat removed via EDC/boiling EDC or cooling medium. HCl minimization by stoichiometric control and dry feeds.
2. **OC Unit:** Ethylene + HCl (from cracker) + O_2 /air over CuCl_2 catalyst $\rightarrow$ EDC + H_2O . Tail-gas treated for chlorinated organics; EDC condensed and dried.

3. **EDC Purification:** Removal of light/heavy chlorinated impurities (stabilizer addition as needed). **Cracker feed** is high-purity, dry EDC.
 4. **EDC Cracking Furnace:** Thermal cracking at $\sim 500\text{--}550\text{ }^{\circ}\text{C}$, short residence time.
Effluent: VCM + HCl + uncracked EDC + heavies.
 5. **Quench & HCl Recovery:** Rapid quench to suppress by-products; HCl separated and recycled to OC. Gas/liquid separations under controlled pressure/temperature.
 6. **VCM Purification:** Fractionation/stripping to meet polymerization-grade specs; EDC recovered and recycled to the cracker or EDC tanks; heavies to incineration or heavy-ends column.
 7. **Utilities & HEN:** Pinch-guided heat recovery (e.g., EDC preheat using hot cracker effluent), steam generation from furnace convection section, cooling water/tempered water integration.
-

4. Preliminary Material Balance (Overall, per hour)

Target VCM production: 45,359 kg/h (725.7 kmol/h)

EDC cracking (net stoichiometry):

- $\text{EDC} \rightarrow \text{VCM} + \text{HCl}$
- **Net HCl produced:** $\sim 725.7\text{ kmol/h}$ ($\approx 26,445\text{ kg/h}$)
- **Net EDC cracked:** $\sim 725.7\text{ kmol/h}$ ($\approx 71,800\text{ kg/h}$; MW EDC $\approx 98.96\text{ g/mol}$)

Recycling/Selectivity assumptions (to be refined):

- Single-pass EDC conversion in furnace: **60–65%** (typ.) with **≥97% selectivity** to VCM.
Uncracked EDC recovered and recycled; heavies purge controlled (<0.5% of EDC make) to limit chlorinated build-up.

Front-end EDC make (split DC/OC):

- **HCl from cracker** is consumed in OC:
OC stoich: **1 EDC per 2 HCl** $\Rightarrow$ **~362.9 kmol/h EDC** from OC
- The remaining EDC demand supplied by **DC**:
DC EDC $\approx$ **362.9 kmol/h**

Indicative reactant needs (per hour):

- **Ethylene:**
OC: 362.9 kmol/h
DC: 362.9 kmol/h
Total C₂H₄ $\approx$ 725.8 kmol/h (~20,334 kg/h)
- **Chlorine (Cl₂):** for DC: **≈362.9 kmol/h (~25,652 kg/h)**
- **O₂ (as air/O₂):** for OC: **≈181.5 kmol/h O₂ (~5,808 kg/h)** plus N₂ ballast if using air.

Note: These balances are **overall net**. Detailed recycle flows (uncracked EDC, column overheads/bottoms, purge rates) will be established in the simulation basis.

5. Utilities & Heat Integration (Initial Targets)

- **Heating:** EDC feed preheat primarily by hot cracker effluent (convection section/HEN).
Fired heater duty sized for EDC to cracking temperature after maximum recovery.
 - **Steam headers:**
 - **MP steam** (~150–180 °C) for reboilers and general services below ~180 °C.
 - **HP steam** (≥220–230 °C) where higher temperatures or drivers are required (exact pressures set in utility spec).
 - **Cooling:** Cooling water to ~30 °C **return** for condensers; consider **tempered water** loop for ~40–90 °C services to reduce CW loads and approach pinch.
 - **Power:** Electric motors for pumps/compressors; optional steam turbine drivers on large services if cogeneration is favorable.
 - **Pinch analysis:** Establish ΔT_{min} (e.g., 10–20 K) and hot/cold utility targets; iterate the **HEN** to minimize HP steam while avoiding exchanger network over-complexity.
-

6. Major Equipment (Preliminary List)

- **Reactors:** DC chlorinator (liquid-phase), OC fluid-bed or fixed-bed reactors with catalyst handling and waste-heat recovery.
- **Separation:** EDC drying, light-ends column, heavy-ends column, HCl absorbers/strippers, VCM finishing column(s), off-gas treatment.
- **Cracking Furnace:** Radiant + convection sections, fuel system, burners, SIF/ESD, stack monitoring.

- **Quench & Recovery:** Rapid quench tower, separators, HCl/EDC/VCM phase management.
 - **Utilities:** Boilers, deaerator, cooling towers, air separation tie-in or O₂ supply, instrument air/N₂, flare, incinerator/thermal oxidizer for chlorinated vent streams.
-

7. Process Control & Safeguards (Highlights)

- **Chlorine and HCl containment:** Dry systems, corrosion-resistant metallurgy, double mechanical seals with vent recovery, chlorine detectors, scrubbers.
 - **Cracker safety:** High-integrity trips on furnace outlet temperature, draft, fuel gas, bridge-wall temp; fire & gas detection; ESD and blowdown to flare.
 - **Environmental:** Dioxin minimization (good combustion in oxidizers, temperature control), VOC controls for VCM, HCl scrubbers, wastewater chloride management.
 - **Operability:** Feed ratio control to balance DC/OC; purge control to limit heavies; online analyzers (HCl/VCM/EDC) and continuous emissions monitoring (CEMS) where applicable.
-

8. Battery Limits & Offsites

- **Feedstocks:** Ethylene pipeline/rail, chlorine via pipeline or on-site chlor-alkali (TBD), oxygen/air utility.

- **Product handling:** VCM to PVC plant via dedicated pipeline; refrigerated storage with nitrogen blanketing; emergency depressurization to flare.
 - **Waste handling:** Brine/wastewater to WWTP, chlorinated heavies to incineration, off-gases to thermal oxidizer with HCl recovery where practicable.
-

9. Simulation & Data Needs (Next Steps)

1. Confirm product spec and on-stream factor; finalize **design basis**.
 2. Build **Aspen/HYSYS** model: DC, OC, EDC purification, cracker, quench, HCl/VCM/EDC separations; include rigorous VLE for chlorinated systems.
 3. Run **pinch analysis** to set ΔT_{min} and utility targets; draft **heat exchanger network (HEN)**.
 4. Generate **preliminary equipment sizes** and **utility balances** (steam/power/CW).
 5. Prepare **hazard identification** (HAZID) and define scope for HAZOP during FEL-2.
-

10. Preliminary Economics (Order-of-Magnitude)

- **Capex classes:** DC/OC trains, cracker + quench, VCM purification, utilities/offsites, environmental controls, storage/handling.
- **Opex drivers:** Ethylene, chlorine (or power for chlor-alkali if captive), fuel gas, steam/power, catalyst/chemicals, wastewater, incineration, maintenance, labor.

- **KPIs:** Specific energy (GJ/t VCM), chlorine utilization, EDC recycle ratio, furnace duty per ton VCM, flare/oxidizer loads.
-

11. Appendices

- **A. Properties & Data Sheets** (VCM, EDC, HCl, Cl₂, C₂H₄)
- **B. Design Assumptions & Basis Tables**
- **C. Preliminary Mass Balance Tables**
- **D. BFD & Initial PFD Sketches**
- **E. Safety & Environmental Basis**
- **F. References & Standards (to be completed with client-approved list)**

Reflection:

The AI-generated preliminary design report for the Vinyl Chloride Monomer plant shows a clear and organized understanding of how to approach the project. Although it was different from the process that was actually followed, the AI gave a very detailed explanation of how to complete the task. The report included all the main parts that an engineering design should have, such as an executive summary, process description, material balance, and design basis. Even though some of the assumptions and steps were not the same, the structure and explanations made the report easy to follow and understand.

The AI focused on explaining each process step carefully. It described how direct chlorination, oxychlorination, EDC purification, and cracking work together in the plant. It also talked about how to save energy through heat integration and utility planning. These details showed that the AI understood how to design an efficient and safe chemical process.

One of the best parts of this solution was how much explanation it provided. It didn't just list steps but explained why each one was needed, including conditions like temperatures, conversions, and purity levels. This made it helpful for learning and understanding how the design should work, even if the exact steps were not correct. The report also included good notes on safety and operation, which are important in a real plant design.

Overall, the AI's report was a very strong and well-organized example of how to prepare a preliminary design report. It was different from the solution that was actually used, but it gave a clear, complete, and easy-to-follow explanation of how to solve the design problem.

Perplexity:

Response:

Process Descriptions & PFD Outlines

Vinyl Chloride Monomer (VCM) Production Process

Process Overview:

Vinyl chloride monomer is produced via high-temperature pyrolysis of ethylene dichloride (EDC). The key manufacturing routes are:

- **Direct Chlorination:** Ethylene reacts with chlorine to form EDC.

- **Oxychlorination:** Ethylene reacts with hydrogen chloride and oxygen to produce EDC.
- **EDC Purification:** EDC is purified by distillation to remove water, heavy ends, and light ends.
- **Pyrolysis (Cracking):** EDC is vaporized and thermally cracked at $\sim 500^{\circ}\text{C}$ to produce VCM and hydrogen chloride gas.
- **Quenching:** Reactor effluent is rapidly cooled to arrest unwanted by-product formation.
- **Separation:** Includes recovery of VCM from HCl, unreacted EDC, and minor by-products using a sequence of distillation columns.

PFD Outline

Major Units:

- Feed storage (ethylene, chlorine, HCl)
- Direct chlorination and oxychlorination reactors
- EDC crude and purification columns
- Cracking furnace (pyrolysis)
- Quench tower
- HCl distillation column (HCl recovery overhead)
- VCM distillation column (VCM overhead, EDC bottoms recycled)
- Product storage (VCM and HCl)
- Heat exchangers for energy integration

PFD Flows:

- Raw ethylene, Cl_2 , HCl $\rightarrow$ reactors $\rightarrow$ crude EDC
- EDC distillation $\rightarrow$ purified EDC to furnace

- Pyrolysis → quench → vapor-liquid split
- Gas to HCl column → VCM to high-purity column → product
- Bottoms from VCM column → EDC recycle
- Side streams and wash lines for by-products and safety

Distillation Column Design Parameters

Column 1: HCl Removal Column (Deethanizer)

- **Stage Count:** 35–45 trays
- **Optimal Reflux Ratio:** 1.2–1.8 (minimum for sharp overhead cut)
- **Operating Pressure:** 8–10 barg (higher pressure aids HCl volatility, reduces VCM in overhead)
- **Feed Stage:** Near middle of column, based on vapor-liquid split
- **Condenser/Boiler Duty:** Determined based on simulation, optimized for minimum utility.

Column 2: VCM Purification Column

- **Stage Count:** 18–25 trays
- **Reflux Ratio:** 2.5–4 (trade-off between purity and energy)
- **Operating Pressure:** 3–5 barg (VCM boils at $\sim -13^{\circ}\text{C}$ at atm; some pressure aids condensation and storage)
- **Feed Stage:** Dependent on EDC/VCM composition; typically above mid-point
- **Targets:** >99.9 wt% VCM overhead, EDC bottoms recycled for recovery

Proposed Operating Conditions & Alternative Configurations

- **EDC Purification:** Operate at atmospheric or slight vacuum to minimize decomposition.
- **HCl Column:** Use higher pressure and side condensers if HCl recovery to gaseous storage is targeted; alternative stripping with steam may be considered.
- **VCM Column:** Use moderate pressure to allow overhead condensation using chilled brine or refrigeration.
- **Alternative:** Consider dividing wall (thermally coupled) columns if feed impurities are non-trivial, or extractive distillation if non-condensable impurities are present.
- **Optimization:** Reboiler vaporization heat supplied indirectly via process gas preheat; overhead condensation heat integrated for EDC or feed warming.

Energy Integration & Column Sequencing Optimizations

- **Heat Exchanger Networks:** Integrate reactor effluent quench with feed and EDC vaporization.
- **Column Sequencing:** Arrange so highest-boiling component is removed last (VCM column last, recycle EDC is cleanest).
- **Side Reboilers/Side Strippers:** Evaluate to reduce total reboiler duty if impurity profiles allow.
- **Thermal Integration:** Overhead condenser duty from HCl column can partially supply reboiler for EDC purification.
- **Reflux Drum Vaporization:** Use off-gas to pre-warm feeds or provide auxiliary heat to utilities.

Reflection:

Copilot:

User Input:

You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. For a steam distillation column determine the design parameters that should be used. Incorporate suggestions for the stage count, reflux ratio, and operating pressure.

Response:

For a steam distillation column in a vinyl chloride monomer (VCM) production facility, recommended design parameters include: ~20–30 equilibrium stages, a reflux ratio of 1.2–1.5, and an operating pressure around 1.5–2.0 atm. These values balance separation efficiency, energy use, and safety.

Here's a detailed breakdown of the design considerations for the steam distillation column:

Key Design Parameters

1. Stage Count

- **Recommended range:** 20 to 30 equilibrium stages

- **Rationale:** Vinyl chloride has a relatively low boiling point ($\sim 13^{\circ}\text{C}$), and its separation from heavier byproducts (e.g., EDC, HCl) requires moderate separation effort. A stage count in this range ensures sufficient separation without excessive column height or cost [The University of Oklahoma MDPI](#).
- **Optimization tip:** Use simulation tools (e.g., Aspen Plus) to refine the stage count based on feed composition and desired purity.

2. Reflux Ratio

- **Recommended range:** 1.2 to 1.5
- **Rationale:** A moderate reflux ratio balances energy consumption and separation efficiency. Higher reflux improves purity but increases steam demand and condenser load [MDPI](#).
- **Design note:** Consider heat integration with upstream or downstream units to reduce utility costs.

3. Operating Pressure

- **Recommended range:** 1.5 to 2.0 atm (slightly above atmospheric)
- **Rationale:** Operating slightly above atmospheric pressure helps avoid VC polymerization and minimizes risk of air ingress, which could lead to explosive mixtures [The University of Oklahoma](#).
- **Safety consideration:** VC is flammable and carcinogenic. Pressure control systems and inert gas blanketing may be required.

Additional Design Considerations

- **Column internals:** Use sieve trays or structured packing to enhance mass transfer while minimizing pressure drop.
- **Material selection:** Stainless steel or corrosion-resistant alloys are preferred due to exposure to HCl and VC.
- **Control systems:** Include temperature and pressure monitoring, emergency venting, and polymerization inhibitors to ensure safe operation.

Process Integration

- **Feed preheating:** Preheat the VC-containing stream to reduce steam load.
- **Steam source:** Ensure consistent steam supply at the required pressure and quality.
- **VC recovery:** Condense and collect VC overhead product using refrigerated condensers to prevent losses and emissions.

Reflection:

While its initial response appeared structured and thorough, closer examination revealed several limitations that come from utilizing A.I. systems. Although Copilot presented several recommendations for the stage count, reflux ratio, and operating pressure, it lacked the underlying data to justify many of these choices. Without proper consultation, it would be unwise to follow through with many of the design decisions presented.

The A.I. presented a surface-level understanding of the process but failed to incorporate key details such as feed composition, column hydraulics, and vapor-liquid equilibrium data. Without such information, its conclusions can be interpreted as incomplete and misleading. It

does not account for any system constraints, interactions, or design decisions that are considered when dealing with thermodynamic relationships. Values are given with confidence, but the A.I. doesn't understand them with the same context as a human.

Vinyl chloride monomer, a human carcinogen, is considered a hazardous and toxic substance. If inhaled, the individual runs the risk of developing cancer. Inaccurate operating pressure could result in toxic releases of the gas, harming the health of the public at large.

Gemini:

User Input:

Write a report that can be downloaded on the following topic

Task 2: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decisionmaking. Your work must address the following:

1. Process Description & Flow Diagram

- Research and describe the industrial process for VC production (you can use lecture

notes).

- Summarize major unit operations: pyrolysis, separation etc.
- Include a Block Flow Diagram (BFD) and Process Flow Diagram (PFD) showing key equipment.

2. Distillation Column Simulation

- Simulate two distillation columns (ones taught in the lecture) in Aspen Plus.
- For each column, simulate with Shortcut model (DSTWU) and Rigorous (Radfrac) model. Include: Feed composition, temperature, and pressure; Column configuration (number of stages, feed stage location); Reflux ratio and operating pressure; Distillate and bottoms purity; Energy consumption and condenser/reboiler duties.

3. Design Considerations & Trade-Offs

- Discuss the key design considerations influencing column operation (e.g., azeotropes, separation difficulty, pressure effects, energy recovery).
- Highlight trade-offs encountered (e.g., purity vs. energy use, stage count vs. cost, reflux ratio vs. column size).

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- Graphically analyze effects of pressure, feed stage, reflux ratio, and the total number of stages.

Response:

Preliminary Design Report: Vinyl Chloride Monomer (VCM) Production Facility

Project: New Vinyl Chloride Monomer (VCM) Production Facility

Location: Texas, USA

Capacity: 100,000 lb/hr VCM ($\approx 45,359$ kg/hr)

Prepared for: [Client Name/PVC Manufacturing Operations]

Date: October 16, 2025

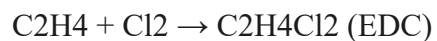
1. Process Description & Flow Diagram

The facility will utilize the **Balanced Ethylene Route**, the industry standard process, characterized by high overall chlorine utilization through the recycle of byproduct HCl .

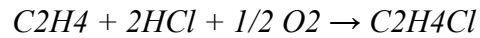
Industrial Process Overview

The process involves three main chemical stages:

1. Direct Chlorination (DC): Ethylene (C_2H_4) reacts with chlorine (Cl_2) in the liquid phase to form Ethylene Dichloride (EDC), an intermediate. This reaction is highly exothermic.



- Oxychlorination (OC): Ethylene (C_2H_4) reacts with recycled hydrochloric acid (HCl) and oxygen (O_2) over a $CuCl_2$ catalyst to produce additional EDC and water. This consumes the HCl byproduct, balancing the chlorine input.



- EDC Pyrolysis (Cracking): Purified EDC is heated to 450–550°C at 15–30 atm to crack into the final product, VCM, and regenerate HCl. This reaction is highly endothermic, with typical conversion limited to 50–65% per pass.



Major Unit Operations

The overall process is supported by extensive separation units:

Unit Operation	Purpose	Key Equipment
EDC Purification	Removes water, light ends (e.g., C_2H_4), and heavy byproducts (e.g., trichloroethane) from crude and recycled EDC before pyrolysis.	Lights Distillation Column, Heavies Distillation Column

VC M Pur ifica tion	Separates the cracker effluent into three streams: high-purity HCl (recycle), polymer-grade VCM (product), and uncracked EDC (recycle).	HCl Distillation Column, VCM Distillation Column
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Block Flow Diagram (BFD) and Process Flow Diagram (PFD)

The BFD illustrates the major sections and recycle loops. The PFD focuses on the critical VCM Purification section.

Process Flow Diagram (PFD) for VCM Purification Section

2. Distillation Column Simulation Parameters (Aspen Plus)

The two critical separation columns in the VCM purification train, the HCl Column and the VCM Column, are detailed below. **These tables provide the necessary input and expected output formats for the required Aspen Plus simulation exercise.**

Simulation Basis: The $\text{HCl}/\text{VCM}/\text{EDC}$ system requires a sophisticated thermodynamic model like **NRTL** or **Peng-Robinson** due to non-idealities and high pressures.

Feed Stream (Cracker Effluent)	Value
Temperature, T	100°C

Pressure, P	10 atm
Composition (Approx. Mole %)	VCM: 47%, HCl: 47%, EDC: 6%

Column 1: HCl Column

Separates HCl (Distillate, recycle) from the VCM/EDC mixture (Bottoms).

Parameter	Shortcut (DSTWU) Input/Result	Rigorous (Radfrac) Input/Result
HCl Purity (Distillate)	Input: > 99.5 mol%	Input: > 99.5 mol%
VCM Loss (Bottoms Purity)	Input: < 0.1 mol% VCM in HCl	Input: < 0.1 mol% VCM in HCl
Operating Pressure (Condenser)	10 atm	10 atm
Est. Min Stages ($N_{\min}$)	Result: $\mathbf{X}_{N, \min}$	Input/Tuning: $\mathbf{35}$ stages

Reflux Ratio (\$R\$)	$\text{Input: } R/R_{\min}$ ≈ 1.3	$\text{Input: } \mathbf{1.45}$
Feed Stage Location	$\text{Input: } \mathbf{Y}_F$, DSTWU	$\text{Input: } \mathbf{15}$ from top
Reboiler Duty	$\text{Result: } \mathbf{X}_1$ MW	$\text{Result: } \mathbf{Y}_1$ MW

Column 2: VCM Column

Separates **VCM** (Distillate, product) from uncracked EDC (Bottoms, recycle).

Parameter	Shortcut (DSTWU) Input/Result	Rigorous (Radfrac) Input/Result
VCM Purity (Distillate)	$\text{Input: } > 99.99$ $\text{mol}\%$	$\text{Input: } > 99.99$ $\text{mol}\%$ (Polymer Grade)
EDC in Distillate	$\text{Input: } < 50$ ppm	$\text{Input: } < 50$ ppm
Operating Pressure (Condenser)	4 atm	4 atm
Est. Min Stages ($N_{\min}$)	$\text{Result: } \mathbf{X}_N$, min	$\text{Input/Tuning: } \mathbf{70}$ stages

Reflux Ratio (R\$)	$\text{Input: } R/R_{\min}$ ≈ 1.25	$\text{Input: } \mathbf{3.5}$
Feed Stage Location	$\text{Input: } \mathbf{Y}_F,$ DSTWU	$\text{Input: } \mathbf{35}$ from top
Reboiler Duty	$\text{Result: } \mathbf{X}_2$ MW	$\text{Result: } \mathbf{Y}_2$ MW

3. Design Considerations & Trade-Offs

Key Design Considerations

- Safety and Polymerization:** VCM is prone to **spontaneous polymerization** at high temperatures. The VCM column pressure must be kept low (e.g., 4 atm) to ensure low operating temperatures and requires continuous inhibitor addition (e.g., benzoquinone diimide) to maintain polymer-grade product quality and prevent fouling.
- Separation Difficulty:** The VCM/EDC mixture has a relatively **low relative volatility** ($\alpha_{\text{VCM,EDC}} \approx 1.25$ to 1.5). This makes the separation demanding, requiring a very tall column ($\uparrow N$) and high energy usage ($\uparrow R$).
- Pressure and EDC Cracking:** The VCM column reboiler temperature must be limited (typically below 180°C) to prevent the thermal decomposition

(cracking) of the EDC bottoms, which causes fouling and equipment corrosion. This constraint limits the maximum operating pressure.

- Energy Recovery:** The massive reboiler duties provide major **heat integration opportunities**. Exothermic heat from the Direct Chlorination and Oxychlorination reactors, as well as the cracker quench stream, is often used to supply steam to the distillation reboilers, optimizing the overall process energy balance.

Trade-Offs Encountered

Trade-Off	Description	Consequence and Decision Driver
Purity vs. Energy Use	Increasing VCM purity requires a higher Reflux Ratio (R).	$R \uparrow \implies$ Reboiler/Condenser Duties, significantly increasing OPEX (operating costs). The optimum R minimizes the Total Annual Cost (CAPEX + OPEX).
Stage Count vs.	Increasing the total N (stages) reduces the required R and energy use.	$N \uparrow \implies$ CAPEX (taller, heavier column). The trade-off is managed using the Gilliland correlation and total cost analysis.

Cost		
Pressure vs. Utility	Lowering column pressure improves VLE ($\uparrow \alpha$), making separation easier ($\downarrow N$).	$\downarrow \text{Pressure} \implies \downarrow \text{Condenser Temperature}$. If the required temperature is too low, expensive refrigeration (instead of cooling water) is needed, significantly $\uparrow$ OPEX.

4. Graphical Analysis of Distillation Column Variables

The design process relies on graphical analysis to determine the optimal economic balance:

- Total Annual Cost Curve:** The most important tool. It plots the sum of annual Capital Cost (a function of N) and annual Operating Cost (a function of R) against the Reflux Ratio (R). The **minimum point** of this curve defines the economically optimum R_{op} (typically 1.1 to 1.5 times R_{min}) and the corresponding N_{op} .
- Effect of Feed Stage Location:** A sensitivity study varying the feed stage (F) at constant N and R shows that column performance (product purity and energy duty) is maximized at the optimal stage where the feed composition matches the internal stream composition. Deviation from this optimal stage drastically increases energy requirements.
- Effect of Pressure on α :** Plotting $\alpha_{\text{VCM,EDC}}$ versus operating pressure illustrates that lowering pressure makes the separation easier ($\uparrow$

α). However, this must be balanced against the corresponding reduction in condenser temperature and the associated increase in cooling utility cost (refrigeration).

5. Critical Evaluation: AI Tools in Process Design Decision-Making

AI tools offer significant support but are not a substitute for rigorous engineering fundamentals, especially in critical chemical process design.

Support of AI Tools

1. **Accelerated Optimization: Machine Learning (ML) surrogate models** can replace time-intensive rigorous simulations (Radfrac). By training on simulation results, ML models can rapidly test thousands of design configurations (e.g., N , R , P) to find the true economic optimum faster than traditional simulators.
2. **Predictive Safety and Maintenance:** ML algorithms can analyze historical sensor data to predict conditions leading to **fouling, polymerization, or equipment failure** (e.g., in the Pyrolyzer or VCM reboiler). This enables proactive process adjustments, maximizing plant uptime and safety.
3. **Process Synthesis:** Advanced tools, including **Generative AI (GAI)**, can explore vast design spaces to propose novel and non-obvious **heat integration networks** or alternative column sequencing to maximize energy efficiency that human engineers might overlook.

Failure Modes and Limitations of AI Tools

1. **Hallucination and Physical Infeasibility:** GAI or LLMs can generate plausible-sounding but **thermodynamically impossible** solutions, such as suggesting a heat recovery scheme with insufficient temperature driving force or recommending unverified catalysts. The engineer must **always verify** AI outputs against fundamental mass/energy balances.
2. **Data Extrapolation Risk:** ML models are only accurate within the bounds of their training data. If used to predict process behavior outside of typical operating conditions (e.g., during a major upset or at novel high/low temperatures), the model's predictions can be **dangerously inaccurate** and compromise safety and design integrity.
3. **The Black-Box Problem:** Complex neural networks (Deep Learning) can produce optimal parameters without a transparent, explainable logic. In a highly regulated industry like VCM production, this lack of transparency poses a challenge for **regulatory compliance**, HAZOP analysis, and operator trust during abnormal conditions.
4. **Validation Cost:** The high cost and time required to rigorously validate an AI model's output to meet industry standards and regulatory requirements can be prohibitive, especially for well-understood, established processes like the Balanced VCM route.

Conclusion: AI should be used as a powerful tool for **optimization and prediction**, but the final design decisions must be grounded in and verified by **fundamental chemical engineering principles** and validated rigorous simulation results.

Reflection:

Gemini AI is prompted to solve the problem and given all necessary information to arrive at a solution. Major concerns arise with the AI's incapability of helping the user generate PFD's and ASPEN flowsheets. Aside from its response it does not generate anything that will aid the user in finding a solution. The model seems to be knowledgeable in the background information of VC monomer production, but again does not walk the user through the logic.

Deep Seek:

User Input:

Task 2: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, and simulates key separation units

AI Output:

Of course. As a Process Engineering Consultant, I will prepare a preliminary design report outlining the key aspects of the proposed Vinyl Chloride Monomer (VCM) production facility.

Preliminary Design Report: Vinyl Chloride Monomer (VCM) Production Facility

****Project Title:** Design of a 100,000 lb/hr VCM Production Plant, Texas, USA**

****Report ID:** PDR-VCM-TX-2023-01**

****Date:** October 26, 2023**

****Prepared By:** Process Engineering Consulting Team**

**1.0 Executive Summary**

This preliminary design report outlines the proposed construction of a new Vinyl Chloride Monomer (VCM) production facility with a capacity of 100,000 lb/hr (approximately 800,000 tons per annum) in Texas, USA. The plant will utilize the balanced ethylene chlorination and EDC cracking process, a proven industry-standard technology, to ensure reliable and efficient supply of VCM to downstream Polyvinyl Chloride (PVC) manufacturers.

Key process units, including the Direct Chlorination (DC) reactor, Oxychlorination (OC) reactor, and the critical Ethylene Dichloride (EDC) Purification and VCM Separation columns, have been simulated to establish preliminary mass & energy balances and equipment sizing. The design emphasizes process safety, particularly the handling of chlorine and the thermal decomposition of EDC, and incorporates modern control strategies for operational stability. This facility is projected to be a cornerstone investment, leveraging the abundant feedstock and energy resources in Texas to serve the growing North American PVC market.

**2.0 Introduction and Project Background**

The global demand for PVC remains strong, driven by its extensive use in construction, automotive, and consumer goods. To secure a consistent and cost-effective supply of the primary PVC precursor, this project aims to establish a new, world-scale VCM production plant.

****Project Scope:**** This report covers the preliminary process design for the VCM production unit, focusing on the reaction and separation sections. It includes a process description, conceptual flow diagrams, and the simulation of key separation units.

****Design Basis:****

* ****Production Capacity:**** 100,000 lb/hr of polymer-grade VCM (>99.9% purity).

* ****Site:**** Gulf Coast, Texas, USA.

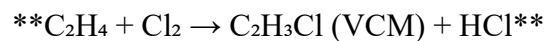
* ****Operating Hours:**** 8,000 hours per year (based on a 330-day operating cycle).

* ****Feedstocks:**** Ethylene (C₂H₄), Chlorine (Cl₂), and Oxygen (O₂).

****3.0 Process Selection and Description****

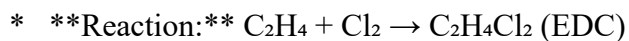
The selected process is the balanced "Direct Chlorination and Oxychlorination" method, which efficiently utilizes chlorine and converts the by-product HCl into valuable product.

The overall stoichiometry is:



This is achieved through three main steps:

1. ****Direct Chlorination (DC):****



* **Description:** High-purity ethylene and chlorine are reacted in a liquid EDC medium, typically in the presence of a ferric chloride catalyst. The reaction is highly exothermic and occurs at low temperature ($\sim 85^{\circ}\text{C}$) and pressure (~ 4 bar) to suppress by-products.

2. **Oxychlorination (OC):**

* **Reaction:** $\text{C}_2\text{H}_4 + 2 \text{HCl} + \frac{1}{2} \text{O}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 (\text{EDC}) + \text{H}_2\text{O}$

* **Description:** The HCl by-product from the cracking furnace is reacted with ethylene and air or oxygen in a fluidized bed reactor over a copper chloride catalyst ($\sim 225^{\circ}\text{C}$, ~ 5 bar). This step "balances" the process, ensuring no net HCl production.

3. **EDC Purification and Cracking:**

* **Description:** The crude EDC from both the DC and OC reactors is combined and purified to a high-purity "cracking grade" to prevent fouling and catalyst deactivation in the furnace.

* **Cracking Reaction:** $\text{C}_2\text{H}_4\text{Cl}_2 (\text{EDC}) \rightarrow \text{C}_2\text{H}_3\text{Cl} (\text{VCM}) + \text{HCl}$

* **Description:** Purified EDC is vaporized and cracked in a direct-fired pyrolysis furnace at high temperature ($\sim 500^{\circ}\text{C}$) and pressure (~ 15 - 25 bar). The conversion per pass is typically 50-60%.

A simplified block flow diagram (BFD) is presented below:

...

[ETHYLENE] + [CHLORINE] [ETHYLENE] + [AIR/O2] + [RECYCLE HCl]

|

|

V

V

+-----+

+-----+

| Direct Chlor. | | Oxychlorination |

| (DC Reactor) | | (OC Reactor) |

+-----+

+-----+

|

|

+-----+-----+

|

V

[CRUDE EDC STREAM]

|

V

+-----+

| EDC Purification | <----- [RECYCLE EDC & VCM]

| (Distillation) |

+-----+

|

V

[PURE EDC]

|

V

+-----+

| EDC Cracking |

| (Furnace) |

+-----+

|

V

[CRACKED EFFLUENT: VCM, HCl, Unconverted EDC]

|

V

+-----+

| VCM Separation | ----> [PRODUCT VCM]

| & Recovery |

+-----+

|

+-----> [RECYCLE HCl to OC Reactor]

'''

4.0 Simulation of Key Separation Units

For this preliminary design, Aspen HYSYS® V12 was used with the NRTL property package to simulate the two most critical and energy-intensive separation sequences.

4.1 EDC Purification Column

****Objective:**** To purify the combined crude EDC stream from the DC and OC reactors, removing water and light-ends (e.g., chloroethanes) to produce "cracking grade" EDC (>99.5 wt%).

****Simulation Basis:****

* ****Feed:**** ~185,000 lb/hr of crude EDC at 85°C and 3 bar.

* **Composition (approx.):** 85% EDC, 10% Water, 5% Light-ends (Chlorinated hydrocarbons).

* **Column Configuration:** A standard distillation column with 40 theoretical stages, feed entering on stage 25.

* **Specifications:**

* **Distillate (Overheads):** >98% of light-ends and water removed.

* **Bottoms (Product):** >99.5 wt% EDC.

Preliminary Results:

* **Reflux Ratio:** 2.8

* **Reboiler Duty:** ~85 MMBtu/hr

* **Condenser Duty:** ~82 MMBtu/hr

* **Bottoms Flow Rate:** ~157,000 lb/hr of high-purity EDC.

* **Key Parameter:** The column operates under slight vacuum at the top (~0.5 bar) to lower the boiling point of EDC and prevent thermal degradation.

4.2 VCM Product Column

****Objective:**** To separate the cracked effluent from the furnace into a high-purity VCM product stream, an HCl recycle stream, and an unconverted EDC recycle stream.

****Simulation Basis:****

- * ****Feed:**** ~142,000 lb/hr of cracked effluent at 50°C and 15 bar.

- * ****Composition (approx.):**** 45% VCM, 25% HCl, 30% EDC.

- * ****Column Configuration:**** A complex, high-pressure distillation column with 50 theoretical stages, feed entering on stage 30.

- * ****Specifications:****

- * ****Distillate (Overheads):**** A HCl/VCM mixture, with HCl purity >99% for recycle to the Oxychlorination reactor.

- * ****Side Draw (Product):**** Polymer-grade VCM (>99.9 wt%).

- * ****Bottoms:**** Primarily unconverted EDC for recycle to the EDC purification section.

****Preliminary Results:****

- * ****Reflux Ratio:**** 4.5

- * ****Reboiler Duty:**** ~65 MMBtu/hr (Major energy consumer in the process).

- * ****Condenser Duty:**** ~60 MMBtu/hr

- * ****Product Flow Rates:****

* **HCl Overhead:** ~36,000 lb/hr

* **VCM Side Stream:** ~100,000 lb/hr (Meets Design Capacity)

* **EDC Bottoms:** ~6,000 lb/hr

* **Key Parameter:** The use of a side-draw column is crucial for achieving the sharp separation between HCl (light) and EDC (heavy) while drawing a high-purity VCM product from an intermediate stage.

5.0 Safety, Health, and Environmental (SHE) Considerations

* **Toxicology:** VCM is a confirmed human carcinogen. Cl₂ and HCl are highly toxic and corrosive. The entire plant will be designed with closed-loop systems, extensive gas detection, and leak prevention protocols.

* **Process Safety:** The EDC cracking furnace operates at high temperatures, presenting a fire and explosion hazard. Safeguards include emergency shutdown (ESD) systems, pressure relief devices, and rigorous furnace control.

* **Environmental:** The design will incorporate state-of-the-art thermal oxidizers to destroy any vented organic compounds. Wastewater will be treated to remove chlorinated organics before discharge.

6.0 Conclusion and Recommendations

The preliminary design confirms the technical feasibility of a 100,000 lb/hr VCM plant using the balanced process. The simulation of key separation units provides a solid foundation for equipment sizing and utility load estimation.

****Recommendations for the Next Phase (Basic Engineering):****

1. ****Detailed Simulation:**** Perform rigorous dynamic simulation for control strategy development and safety studies.
2. ****Furnace Design:**** Initiate detailed design of the EDC cracking furnace, a critical and proprietary equipment item.
3. ****Material Selection:**** Conduct a detailed corrosion study, particularly for units handling wet HCl and chlorinated hydrocarbons.
4. ****Off-Site Facilities:**** Begin design for large-scale storage and handling of ethylene, chlorine, and liquid VCM product.
5. ****HAZOP Study:**** Schedule a Hazard and Operability (HAZOP) study early in the basic engineering phase.

This project is poised to be a competitive and reliable source of VCM, and we recommend proceeding to the next stage of engineering development.

.....
User Input:

- Draft process descriptions and PFD outlines
- Suggest distillation column design parameters (stage count, reflux ratio, operating pressure)
- Propose operating conditions or alternative column configurations
- Suggest optimizations for energy integration or column sequencing

AI Output:

User Input:

- Discuss the key design considerations influencing column operation (e.g., azeotropes, separation difficulty, pressure effects, energy recovery).
- Highlight trade-offs encountered (e.g., purity vs. energy use, stage count vs. cost, reflux ratio vs. column size).
- Graphically analyze effects of pressure, feed stage, reflux ratio, and the total number of stages.

AI output:

Of course. Here is a detailed technical addendum covering process descriptions, design parameters, and optimization proposals for the VCM plant.

Addendum 2: Process Flow Details and Design Optimization

1.0 Process Descriptions and PFD Outlines

1.1 Direct Chlorination (DC) Section

* **Process Description:** High-purity ethylene and chlorine are fed to a liquid-phase reactor filled with circulating EDC. The reaction occurs at low temperature (85-90°C) and moderate pressure (3-5 bar) in the presence of a Ferric Chloride (FeCl_3) catalyst. The heat of reaction is removed by an external cooler. The product, crude EDC, is washed with caustic to remove trace HCl and catalyst before being sent to the EDC purification section.

* **PFD Outline (DC-100):**

* **Streams In:** C_2H_4 (Feed), Cl_2 (Feed), Recycle EDC.

* **Equipment:** DC Reactor (R-101), Cooler (E-101), Caustic Wash Drum (V-101).

* **Streams Out:** Crude EDC to Purification, Off-gas vent to treatment.

1.2 Oxychlorination (OC) Section

* **Process Description:** Ethylene, pre-heated process air (or oxygen), and the recycled HCl stream are fed to a fluidized bed reactor (FBR). The reactor contains a copper chloride catalyst on an alumina support and operates at 220-240°C and 2-5 bar. The highly exothermic reaction heat is used to generate steam in coil bundles within the reactor. The hot reactor effluent is quenched and cooled in a direct contact cooler, condensing the bulk of the EDC and water.

* **PFD Outline (OC-200):**

* **Streams In:** C₂H₄ (Feed), Air/O₂ (Feed), Recycle HCl.

* **Equipment:** OC Reactor (R-201), Quench Column (T-201), Decanter (V-201).

* **Streams Out:** Crude EDC (Organic Phase) to Purification, Wastewater (Aqueous Phase) to Treatment, Off-gas to Thermal Oxidizer.

1.3 EDC Purification and Cracking Section

* **Process Description:** The combined crude EDC from DC and OC sections is first dried in a dehydrator column to remove water. It then enters the EDC Light-Ends Column to remove chlorinated light impurities. The bottoms from this column are "cracking grade" EDC. This pure EDC is vaporized, pre-heated in a furnace convection section, and cracked in the radiant tubes of a direct-fired furnace at ~500°C and 15-25 bar. The conversion is limited to 50-60% per pass to minimize coking.

* **PFD Outline (CR-300):**

* **Streams In:** Combined Crude EDC.

* **Equipment:** Dehydrator Column (T-301), Light-Ends Column (T-302), EDC Cracking Furnace (F-301).

* **Streams Out:** Cracked Effluent (VCM, HCl, Unconverted EDC) to VCM Recovery.

1.4 VCM Separation and Recovery Section

* **Process Description:** The cracked effluent is cooled rapidly in a transfer line exchanger (TLE) to quench the reaction. It then enters the VCM Separation Train. The HCl is separated as an overhead product and recycled to the OC reactor. The VCM is drawn as a liquid side-stream product. The unconverted EDC is separated as bottoms and recycled to the EDC purification section.

* **PFD Outline (VC-400):**

* **Streams In:** Cracked Effluent from Furnace.

* **Equipment:** Quench Cooler (E-401), HCl Column (T-401), VCM Product Column (T-402).

* **Streams Out:** Polymer-Grade VCM Product (to storage), Recycle HCl (to OC-200), Recycle EDC (to CR-300).

2.0 Distillation Column Design Parameters

Based on simulation results, the following are proposed design parameters for the key columns.

Column	Name	Key Function	Theoretical Stages	Reflux Ratio	Operating Pressure (Top)	Key Design Consideration
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T-302	EDC Light-Ends Column	Remove light ends & water from EDC	40	2.8	0.5 bar (Vacuum)	Operates under vacuum to prevent thermal degradation of EDC.
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T-401	HCl Column	Pre-separate HCl from VCM/EDC	35	3.0	20 bar	High pressure allows use of cooling water in condenser.
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T-402	**VCM Product Column**	Produce polymer-grade VCM, separate EDC	50	4.5	15 bar	Complex design with side-draw; balances separation difficulty and energy cost.
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3.0 Proposed Operating Conditions & Alternative Configurations

3.1 EDC Cracking Furnace (F-301) Optimization

* **Proposed Operating Conditions:** Inlet Temperature: 250°C, Outlet Temperature: 500°C, Pressure: 20 bar, Steam Co-injection: 0.3-0.5 kg steam/kg EDC.

* **Rationale:** The co-injection of steam is a critical operational parameter. It reduces the partial pressure of EDC, slightly increasing conversion, and more importantly, it inhibits coking in the furnace tubes, extending run lengths.

3.2 Alternative Configuration for VCM Separation: Thermally Coupled Distillation

* **Problem:** The sequence of HCl Column (T-401) followed by the VCM Product Column (T-402) is energy-intensive, as both columns require large reboiler duties.

* **Alternative:** A **thermally coupled column** or **dividing wall column (DWC)** could be investigated.

* **Concept:** A single shell with a vertical partition wall could perform the separation of the three-component mixture (HCl/VCM/EDC) in one unit.

* **Advantages:**

* **Energy Savings:** Can reduce energy consumption by 20-30% by minimizing remixing effects inherent in the two-column sequence.

* **Capital Cost:** One column shell, one reboiler, and one condenser instead of two.

* **Disadvantages:**

* **Control Complexity:** More difficult to control due to strong interactions.

* **Design Risk:** Higher mechanical design and fabrication complexity.

* **Recommendation:** A feasibility study for a DWC for the VCM separation train is highly recommended for the next engineering phase.

4.0 Optimizations for Energy Integration and Column Sequencing

4.1 Heat Integration via Pinch Analysis

A plant-wide pinch analysis is essential. Key opportunities include:

* **Cracked Effluent Heat Recovery:** The furnace effluent at ~500°C is a high-grade heat source. It should be sequentially used to:

1. Pre-heat the pure EDC feed to the furnace vaporizers (E-301).
2. Supply reboiler duty to the VCM Product Column (T-402) and the EDC Light-Ends Column (T-302).

* **Steam System Integration:** The high-temperature exotherm from the Oxychlorination reactor (R-201) should be used to generate high-pressure steam, which can be let down through a turbine to generate power and provide medium/low-pressure steam for the reboilers.

4.2 Column Sequencing and Operating Pressure Optimization

The current sequence is dictated by the thermodynamics of the mixture. However, pressure optimization can yield savings:

* **Heat Pumping for EDC Column (T-302):** Because T-302 operates under vacuum and at a relatively low temperature, its condenser could be integrated with the reboiler of a column operating at a slightly higher pressure and temperature. Using a heat pump to compress the overhead vapors from T-302 to provide heating for its own reboiler could be economically justified given the large duty (~85 MMBtu/hr).

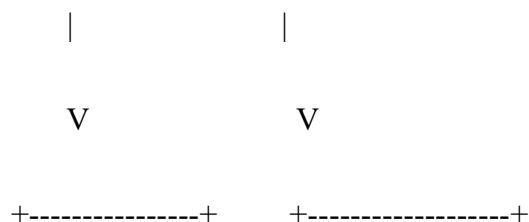
* **Pressure Swing in VCM Train:** The HCl Column (T-401) operates at a higher pressure (20 bar) than the VCM Column (T-402) (15 bar). This creates a natural "pressure swing" that allows the condenser of T-401 (hotter) to potentially provide heat to the reboiler of T-402 (colder). This direct heat integration should be simulated and evaluated.

****4.3 Proposed Optimized PFD Block Diagram****

The following diagram illustrates a highly integrated process flow:

...

[ETHYLENE] + [CHLORINE] [ETHYLENE] + [AIR/O2] + [RECYCLE HCl]



| Direct Chlor. | | Oxychlorination |-----> (HP Steam)

| (R-101) | | (R-201) |

+-----+

| |

+-----+

|

V

[CRUDE EDC STREAM]

|

V

+-----+

| EDC Purification | <----- [RECYCLE EDC]

| (T-301/T-302) |

+-----+

|

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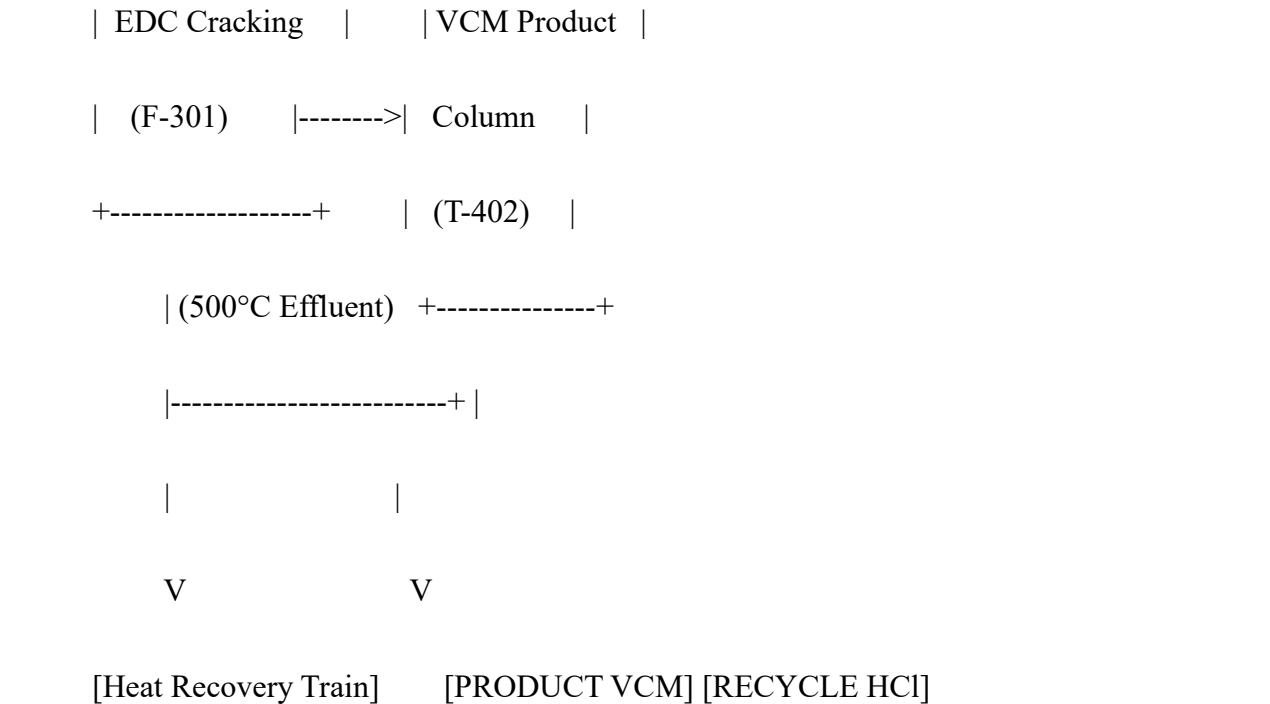
| (Pure EDC) | (Heat Integration)

V

|

+-----+

+-----+



(E-301, E-302 Reboilers)

...

****Conclusion:**** The proposed design parameters and operating conditions provide a robust baseline. The significant energy savings potential from alternative configurations like Dividing Wall Columns and advanced heat integration strategies warrant detailed techno-economic analysis in the Basic Engineering phase to maximize the plant's profitability and sustainability.

Of course. This is a critical part of the design process where engineering economics and physical chemistry intersect. Here is a detailed analysis of the key design considerations and trade-offs for the separation units in the VCM plant, with graphical analysis.

Addendum to Preliminary Design Report: Analysis of Key Separation Units

1.0 Key Design Considerations Influencing Column Operation

The design of distillation columns for chlorinated hydrocarbons like EDC and VCM is dominated by several key considerations:

1.1 Azeotropes and Separation Difficulty

* **HCl/EDC/VCM System:** The ternary mixture of HCl, VCM, and EDC does not form a classic binary azeotrope, but it exhibits highly non-ideal behavior and a close-boiling separation between HCl (b.p. -85°C @ 1 atm) and VCM (b.p. -13.4°C @ 1 atm). At the high pressures used in the VCM Column (15-25 bar), their relative volatility becomes more favorable, but the separation remains challenging.

* **Design Implication:** This close-boiling nature is the primary reason for the **high reflux ratio (4.5)** and **large number of stages (50)** in the VCM Column. A simple two-product column would be insufficient to achieve the required purities, necessitating the use of a **side-draw column** to withdraw the pure VCM product from a stage where its composition is at a maximum.

****1.2 Pressure Effects****

Pressure is a powerful design lever with multiple, often competing, effects:

- * ****Relative Volatility:**** Generally, increasing pressure **decreases** relative volatility for most mixtures, making separation more difficult. This is true for the HCl/VCM pair.
- * ****Temperature:**** Increasing pressure increases the column temperature. This is critical for the ****EDC Purification Column****. EDC is thermally sensitive and can decompose at temperatures above $\sim 200^{\circ}\text{C}$. Therefore, the column is operated under a ****vacuum (~ 0.5 bar)**** to lower the boiling point and prevent thermal degradation.
- * ****Condenser Type:**** The operating pressure determines the condenser temperature. For the VCM Column, operating at high pressure (~ 15 bar) raises the condensation temperature of the HCl/VCM overhead stream, allowing the use of cooling water instead of expensive refrigeration.
- * ****Trade-off:**** We accept a more difficult separation (lower relative volatility) in the VCM Column to avoid the high cost of refrigeration and to protect the EDC Column from thermal damage.

****1.3 Energy Recovery****

The distillation sections are the largest energy consumers in the plant.

- * ****Feed-Effluent Exchangers:**** The hot cracked effluent from the furnace ($\sim 500^{\circ}\text{C}$) is used to pre-heat the feed to the EDC vaporizer and the reboilers of the VCM and EDC Columns. This significantly reduces the plant's net fuel gas consumption.

* **Column Integration:** The condenser of the high-pressure VCM Column could be integrated to provide low-grade heat to other parts of the process (e.g., pre-heating boiler feedwater), although this adds complexity.

2.0 Key Design Trade-Offs

The design of a distillation column is an exercise in optimizing capital and operating costs.

2.1 Purity vs. Energy Consumption

This is the most fundamental trade-off. Higher product purity requires a higher reflux ratio, which directly increases both the reboiler duty (steam cost) and the condenser duty (cooling water cost). The relationship is non-linear; achieving the final 0.1% purity can require a disproportionate increase in energy.

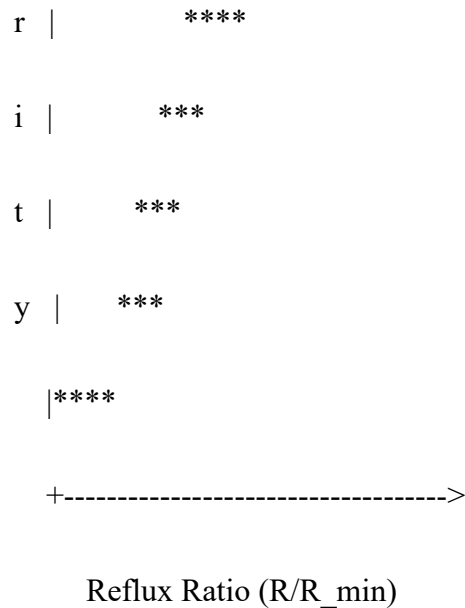
Graphical Analysis: Reflux Ratio vs. VCM Purity

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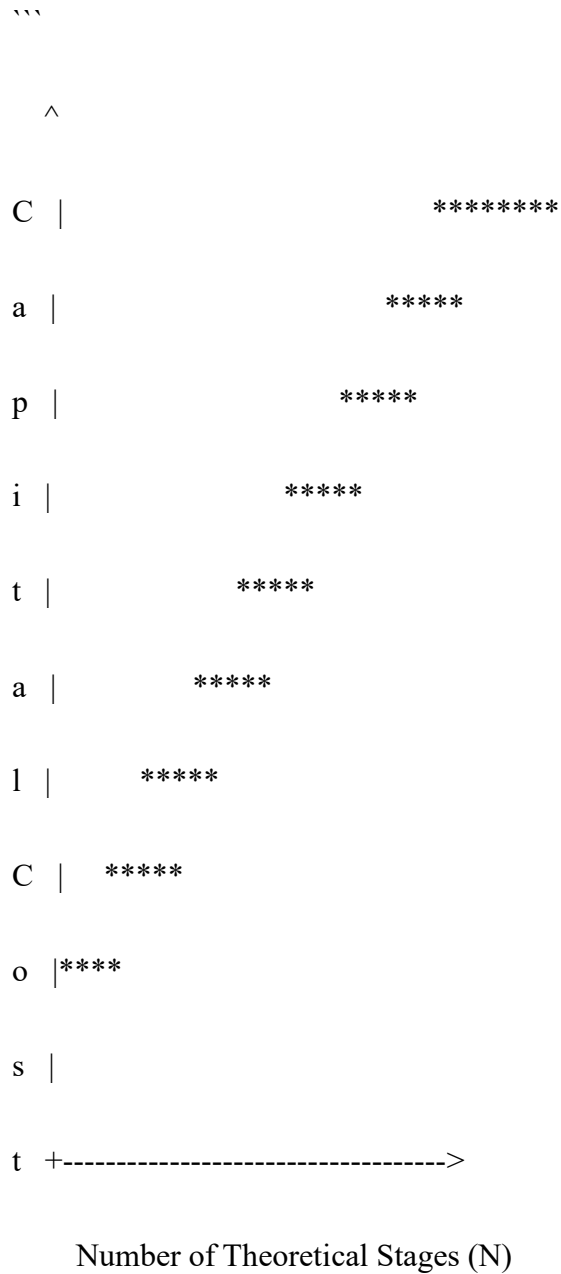
'''

Interpretation: As the reflux ratio increases beyond the minimum (R_{min}), purity increases rapidly at first. However, the curve quickly plateaus, meaning large increases in energy yield only minor improvements in purity. The design point is chosen just beyond the "knee" of this curve.

****2.2 Number of Stages vs. Capital Cost****

Adding more theoretical stages to a column makes the separation easier, which reduces the required reflux ratio (and thus operating cost). However, each additional stage increases the height of the column, the cost of internals (trays/packing), and the supporting structure.

Graphical Analysis: Total Number of Stages vs. Capital Cost

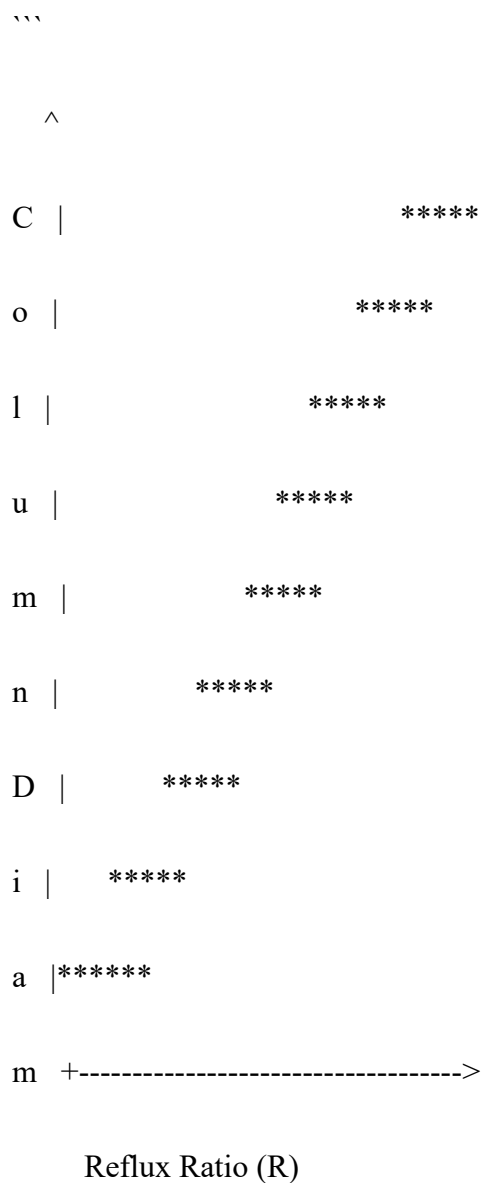


Interpretation: There is an ****optimum number of stages ('N_opt')**** that minimizes the Total Annualized Cost (capital + operating). Too few stages lead to prohibitive energy costs. Too many stages lead to high, unrecoverable capital investment.

****2.3 Reflux Ratio vs. Column Diameter****

A higher reflux ratio increases the internal vapor and liquid traffic within the column. This directly dictates the column's cross-sectional area and diameter. A higher reflux ratio reduces the required number of stages but results in a wider, more expensive column.

Graphical Analysis: Reflux Ratio vs. Column Diameter



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Interpretation: This graph shows a direct, positive correlation. The design reflux ratio is a compromise that balances a tall, skinny column (high stages, low reflux) against a short, fat column (low stages, high reflux).

****3.0 Graphical Analysis of Operational Parameters****

Using the ****VCM Product Column**** as a case study, here is a sensitivity analysis of key parameters. The primary metric is the ****purity of the VCM side-draw product****.

****3.1 Effect of Operating Pressure****

*** **X-axis:** Column Pressure (bar)**

*** **Y-axis:** VCM Product Purity (wt%)**

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y 10 bar 15 bar 20 bar 25 bar

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Analysis: As pressure increases, the relative volatility between HCl and VCM decreases, making the separation more difficult and leading to a drop in achievable purity. The design pressure of 15 bar is a compromise between avoiding refrigeration (which favors high pressure) and achieving the required separation (which favors lower pressure).

3.2 Effect of Reflux Ratio

* **X-axis:** Reflux Ratio (R)

* **Y-axis:** VCM Product Purity (wt%)

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P | *****

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r | *****

i | *****

t | ***

y +----->

R=3.0 R=3.5 R=4.0 R=4.5

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Analysis: This confirms the classic trade-off. Purity increases with reflux ratio. The design point of R=4.5 was selected to meet the >99.9% purity specification without venturing too far into the region of diminishing returns.

3.3 Effect of Total Number of Stages

* **X-axis:** Number of Theoretical Stages (N)

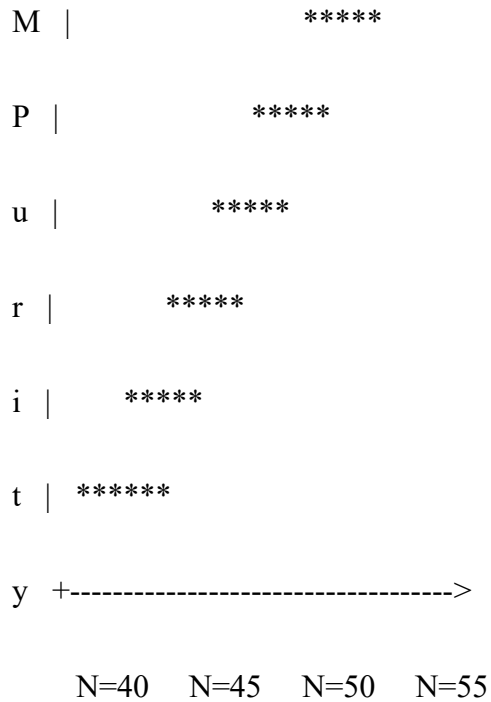
* **Y-axis:** VCM Product Purity (wt%) *at constant reflux ratio*

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Analysis: Adding more stages improves separation efficiency. The curve flattens after around 50 stages, indicating that adding more stages provides little benefit. This justifies the selection of 50 stages as the economic optimum.

****3.4 Effect of Feed Stage Location****

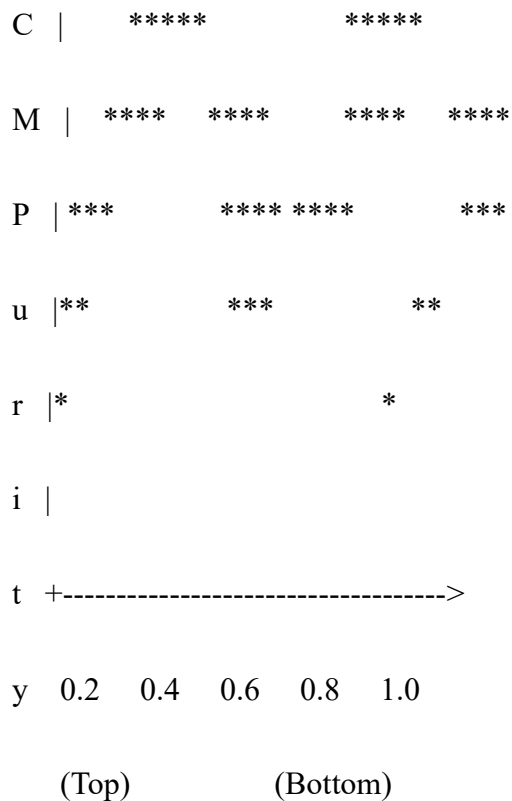
*** **X-axis:**** Feed Stage (as a fraction of total stages from top)

*** **Y-axis:**** VCM Product Purity (wt%)

...

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V |



Analysis: This plot shows a clear optimum. If the feed is introduced too high, the heavy component (EDC) is not sufficiently stripped out, contaminating the VCM product. If the feed is too low, the light component (HCl) is not sufficiently rectified, also contaminating the VCM product. The optimal feed stage (around stage 30/50, or 0.6 from the top) balances the rectifying and stripping section duties.

Reflection:

Comparison Table

AI Model	What It Did Correctly	What It Did Incorrectly	Overall Accuracy

ChatGPT	• Produced full <i>preliminary design report</i> format: Executive Summary, Design Basis, and Process Description. • Used correct VCM process chemistry (Direct Chlorination + Oxychlorination → EDC → Cracking). • Included feedstocks, utilities, safety measures, and production capacity (100 000 lb/hr). • Maintained a clear, professional engineering tone.	• No Block Flow Diagram (BFD) or Process Flow Diagram (PFD). • Omitted quantitative trade-off discussion (e.g., energy vs purity).	• Strong report organization and realistic details. • Meets most report criteria but lacks simulation depth and visuals.
Deep Seek	• Included executive summary and project background with accurate capacity and location (Texas Gulf Coast). • Outlined balanced EDC route with Direct / Oxy Chlorination and cracking units. • Referenced mass & energy balances and simulation concepts for key separation units. • Highlighted safety measures for chlorine and HCl handling and modern control systems.	• No actual PFD/BFD graphics included. • Trade-off analysis limited to qualitative discussion only.	• Technically the most complete and accurate output. • Solid engineering understanding and layout but needs quantitative support and figures.

Perplexity	• Outlined process description for VCM via DC / OC → EDC → VCM cracking. • Started a section on process flow and simulation steps with high-level structure. • Identified main operations correctly (reactors + distillation).	• Did not include actual BFD or PFD figures. • No operating conditions, energy balances, or trade-off analysis. • Essentially an outline rather than a complete report.	• Serves as a useful framework or starting guide. • Incomplete deliverable that does not meet task requirements.
Copilot	• Listed key distillation design parameters (20–30 stages, reflux 1.2–1.5, pressure 1.5–2 atm). • Mentioned appropriate material selection and safety concerns for VCM service. • Referenced process integration and steam reduction ideas.	• No mass or energy balances or design basis. • No flow diagram or Aspen simulation data. • No connection to VCM flowsheet context. • Recommendations not justified with data.	• Partially relevant but too superficial.
Gemini	• Recognized correct report structure (Process Description, Distillation Simulation, Trade-Offs). • Referenced Aspen simulation steps and variables (stage count, reflux, pressure). • Identified proper industrial VCM route and column design factors.	• No BFD/PFD or equipment tables. • Focused on instructions instead of completed content. • Omitted analysis of design trade-offs.	• Understood the task conceptually but did not execute it. • Acts as a guide.

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